

Endpoint Catalogue Service - Requirements



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Glossary of Terms

Term / Abbreviation	What it stands for
	Booking and Referral Standards
BaRS	The Booking and Referral Standard (BaRS) is an interoperability standard that enables digital patient journeys or operational processes for patients, healthcare workers and carers. It allows relevant information to be shared quickly, safely and in a format that is useful. It will eventually be available in all care settings.
BSA	Business Service Authority
Business Identifier	A unique globally available and understood entity broken down into individual entities for a specific service. For example: - ASID - Target identifier - url with specified interaction(s)
CAS	Clinical Assessment Services
CPM	Connecting Party Manager
DoS	Directory of Services
DoS Service ID	Is a unique identifier on the UEC Directory of Services (DoS) for a specific Healthcare Service
Endpoint	A physical address to support a technical transport operation. For example: - ITK - BaRS
ED	Emergency Department
Endpoint Administrator	Anyone deploying or managing the endpoint, for example: - Suppliers - DoS leads - Healthcare Service provider - BaRS Team.
Endpoint Template	A globally defined template for the creation and management of endpoints These templates will be used when configuring a Healthcare service based on a digital service (supplier) providing the solution used by the service. If a Healthcare service is configured with a Product ID where the supplier has defined the template, then that template must be used by the service. Each template will contain as a minimum a Product ID and connection type and address (url) of the service.
EPCSe	Endpoint Catalogue Service
Federated Endpoint	A federated endpoint is a single url (address) for all users of a supplier's solution

Healthcare Service	A Healthcare Service is provided by an Organisation, examples include hospital clinics for elective care, emergency departments, pharmacy services, non-NHS Services or private services.
ICT	Information and Communications Technology
ITK	Interoperability Toolkit The Interoperability Toolkit (ITK) is a set of common specifications, frameworks and implementation guides. It supports interoperability within local organisations and across local health and social care communities.
On-Prem	Short name for On Premise (see illustration at Appendix 5) A solution that is managed and deployed by an Organisation or Healthcare Service rather than by a System Supplier, for example they will manage the security, operating systems, networking, etc.
Organisation	There are different flavours of organisations - Parent for example Boots plc would be parent Boots Leeds. South Central Ambulance Trust would be the parent of Andover Ambulance Station - Managing organisation could be the supplier, trust, GP, etc. - Non-NHS for example opticians, dentists, private healthcare providers
Private Endpoint Template	This template can only be viewed and/or managed by - Own organisation / Healthcare Service - System Supplier for the Organisation / Healthcare Service - DoS Team - NHSE Programmes and Support Teams - NHS Service Desk - NHS Incident Management Service
Product ID	Is a unique identifier applied to the supplier solution through the digital onboarding service
RBAC	Role Based Access Controls
Supplier / System Supplier	The supplier is the company or organisation that provides ICT hardware, software and services to an organisation
UEC	Urgent Emergency Care
URL	Uniform Resource Locator, also called a web address, is a unique identifier used to locate a resource on the internet.

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1.0 Executive Summary

The Endpoint Catalogue Service (EPCSe) is required to facilitate the routing of standard-compliant messages (such as BaRS, GP Connect, or ITK) within the NHS by providing a mechanism to look up and use correct digital endpoints for Healthcare Services. The EPCSe provides for a more flexible, reliable, and robust management of Healthcare Services and the systems that support them through linkage of business identifiers and the digital endpoints.

Key use cases include BaRS (Booking and Referral Standard) and GP Connect, which require the traffic to be routed to a destination based on a business identifier with the ability to decentralise the management of these endpoints from NHSE. Other use cases also include Urgent and Emergency Care (UEC); however, the service could be used in any scenario which requires management of digital endpoints that support Healthcare Services.

The service should operate using an API-first approach, emphasising integration with NHS systems via accessible APIs. It offers a self-service model, enabling users to manage endpoints, Healthcare Services, and their relationships in a decentralised fashion. The data model supports identifying and organising Healthcare Services and their digital endpoints through unique identifiers, ensuring accuracy and flexibility in routing system-to-system communication.

The requirements outlined in this document include functional aspects like registering or managing and retrieving the endpoints and Healthcare Services together with their managing organisations, linking them, as well as maintaining the data integrity. The non-functional requirements focus on typical concerns such as security, performance, compliance with NHS technical and data standards.

The solution will also support role-based access control (RBAC) at the application level for secure management of endpoints and services, with distinct roles assigned for administration and consumer access.

The solution will not however address any issues with the existing or future businesses processes and should be regarded as an agnostic tool that will help improve, streamline and automate those processes.

2.0 Overview

The purpose of the EPCSe is to allow management of the data necessary, for the routing of standard-compliant (such as BaRS-compliant, or ITK-compliant) messages to the required destination. The sending system¹ need only be aware of the business identifier of the Healthcare Service it wishes to reach. The EPCSe will mostly be used (either by the sending system or an intermediary service²) as a look-up service to find the correct address (URL) of a standard-compliant endpoint where the message should be forwarded to. This separation of business identifiers and endpoints is necessary to allow independent definition, and the management, of Healthcare Services and the digital systems that support those services.

2.1 Problem Statement(s)

2.1.1 BaRS

Currently, from a BaRS perspective the DoS endpoint management is a manual process that is managed by BaRS in conjunction with UEC DoS team to facilitate the routing of standard-compliant messages between suppliers and Healthcare Services. The process is prone to error and delays when new endpoints are added, updated or retired. Currently BaRS manages endpoints for pharmacy services and other Healthcare Services for example Northwest NHS 111 service but as BaRS take-up is rolled out to wider Healthcare Services the current process will be unsustainable and prone to more mistakes, which may have significant clinical and business repercussions.

2.1.2 UEC DoS

Currently, the DoS acts as both a service discovery tool and a repository for endpoint addresses for example Pharmacy, Emergency Departments (ED) and Clinical Assessment Services (CAS), etc. (i.e., where referrals are actually sent). In the future, this capability will be available for other NHS and non-NHS Organisations. This dual purpose introduces a range of operational challenges:

Delayed & Inconsistent Updates: in the context of BaRS, changes to pharmacy a system supplier can occur at any time (e.g., 9am), however these changes may not be added to the BaRS endpoint catalogue until a later time (e.g., 4pm). This delay risks out-of-date endpoint information being used for patient referrals via BaRS in the interim.

Duplicated Data Entry: Under the current setup, endpoint information must be managed in multiple systems (DoS and BaRS endpoint catalogue), increasing the mis-matched data. As use cases for BaRS increase, and whilst BaRS is not fully adopted and some systems continue to use ITK, there is a requirement for both systems to hold endpoint information.

Increased Potential for Manual Errors: As endpoint information requires manual updates, the process is prone to human error, creating potential safety and operational risks.

¹ 'sending system' is one that has been assured and approved to utilise digital services provided by NHSE and in the context of this document, for example complaint with BaRS or ITK.

² 'intermediary service' is one that has been provided for integration purposes for non-compliant receiving systems.

Responsibility for Updating Supplier Information: DoS teams are currently tasked with updating system supplier information changes. Ideally, the responsibility for this should sit with the system suppliers themselves, to ensure real-time accuracy and to remove this administrative burden from DoS teams. Although assurance will still need provided to DoS teams for the purposes of issues resolution or queries.

Clinical and Operational Safety Risks: For services to remain safe and effective, referral details must be sent to the correct service/system/provider. Delays, duplication, and manual steps in the current process may compromise patient safety and overall confidence in the service.

In summary, the requirement is to streamline endpoint management by removing endpoint addresses from the DoS and consolidating them into a dedicated EPCSe. This will create a single source of truth for all endpoint look-ups, reducing the risk of manual errors, remove administrative burdens from DoS teams, and ensure clinically safe referrals with up-to-date supplier information.

2.1.3 GP Connect

Not stated

2.2 Use Cases

2.2.1 BaRS

BaRS has a need for routing traffic to the intended destination ascertained by the use of a business identifier. The business identifier will have a related endpoint specifically for use with BaRS.

There is also a need to decentralise the management of these endpoints by allowing management outside of the BaRS Team.

More information about the BaRS business processes/workflow, as-is and to-be can be found at Appendix 3 of this document.

2.2.2 UEC DoS

UEC DoS (Urgent and Emergency Care Directory of Services) presents endpoints to allow point-to-point communication between any two parties. NHS England doesn't broker these requests. However, there is a similar need to robustly manage relationships between sender (e.g. GP practice, 111) and receivers (e.g. pharmacy). Furthermore, DoS users will need to have the capability to receive and read details in the new solution, so that they can convey this on to others, within the context of their Rest or Check Capacity Summary (CCS) APIs that provide end point information to third parties, such as GP systems and 111 provider systems. In this case when a Pharmacy service changes from one supplier to another, their endpoint will change in DoS. It would be very beneficial if CPM (Connecting Party Manager) could be expanded, or a new lightweight service stood up, to manage other endpoint registries and not just those from API-M or SDS.

More information about the UEC DoS business processes/workflow, as-is can be found at Appendix 4 of this document.

2.2.3 GP Connect

The aim is that the consumer system is given a “one stop shop” with everything hidden behind the APIM Proxy.

The consumer/sender system will know what kind of transaction or lookup they need to perform and which target organisation. Currently this is the GP Principal Clinical System (PCS) but that may differ in future use cases. The consumer system is configured with the [Spine Interaction IDs](#) for the API functions they require. Going forward it would be beneficial for organisations to specify which GP Connects products are in use across GP Practices or pharmacies, and where applicable other Healthcare settings.

There may be future use cases where the consumer system knows only what service they require, not which organisation provides it. This might be when they wish to send some data but do not know which org manages the “system of record” of the data. In this case the lookup might be just to a named business service and the target organisation is hidden.

Guiding principles

- The consumer is hidden as far as possible from the complexities of implementation
- The consumer is agnostic of the supplier of the system in use at the target organisation i.e. where the information has either been created or consumed.

MVP Requirements

Healthcare Service and target Organisation IDs, return FHIR-compliant URL.

Authentication/Authorisation: Application-restricted

Nice to have requirements

Multiple responses with an order for preference (e.g. fallback to Send Document/MESH mailbox, or an organisation email address)

Retrieve an endpoint by service only (e.g. when the organisation is unknown).

More information about the GP Connect business processes/workflow, as-is and to-be can be found at Appendix 2 of this document.

2.3 Use Case Diagram and Persona Stories (Scenarios)

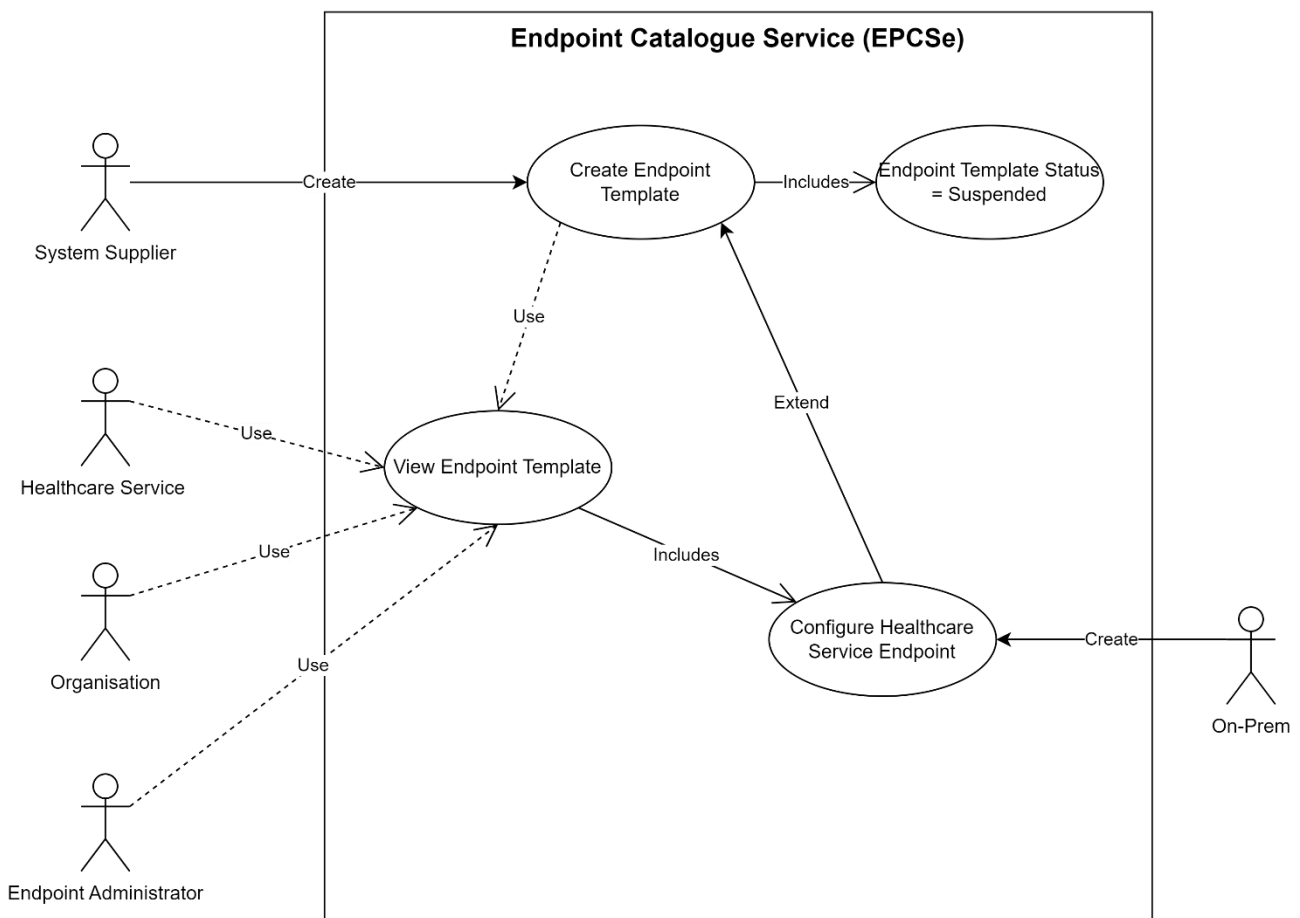


Figure 1: EPCSe Use Case Diagram

2.3.1 Scenario 1: Single Solution for a Single Connection Type

MedRec are a Healthcare Service system supplier for a single solution for a single data sharing method for example BaRS, ITK and a single federated endpoint.

Information regarding MedRecs solution, namely the connection type and address (url) are visible and applied to their endpoint template in the EPCSe. This will enable users within Healthcare Services supplied by MedRec to find and configure the MedRed endpoint.

Liam works for the Swan Lake Pharmacy. He is the organisation's Endpoint Administrator and has been assigned permissions to maintain and manage his organisations Healthcare Services configuration on the EPCSe. This is to ensure the pharmacy can check the Healthcare Service configuration. Liam on accessing EPCSe can view and select the MedRec solution to configure using the defined template.

2.3.2 Scenario 2: Single Solution with Multiple Connection Types and Multiple (Federated) Endpoints

SunHealth are a Healthcare Service system supplier with a single solution that has multiple connection types and multiple (federated) endpoints e.g. the solution (product) incorporates functionality for BaRS, ITK etc, including solution version differences to support breaking changes.

Information regarding Sunhealth's solution is configured in multiple templates for discrete connection types and endpoints, supporting their federated deployment model to allow Healthcare Services to configure their service endpoints. These templates are managed by SunHealth but can be selected by the Healthcare Service.

However, in this instance there will multiple templates (one for each connection type and version of the deployed solution). This will enable users within Healthcare Services supplied by SunHealth to find and configure the appropriate SunHealth endpoint, from the relevant templates.

Trevor works for Oakland Pharmacy. He is the organisation's **Endpoint Administrator** and has been assigned permissions to maintain and manage his organisations Healthcare Services configuration on the EPCSe. This is to ensure the pharmacy can check the Healthcare Service configuration.

Trevor needs to configure his Healthcare Services endpoint to align with the new beta version of SunHealth's solution. On accessing EPCSe he can see two SunHealth templates with different connection types and is able to select the relevant one for the new version (beta) to configure the endpoint.

2.3.3 Scenario 3: Single Solution for a Single Connection Type and Single (On-prem) Endpoint

Jayne works for the Swan Lake Emergency Department, part of Somerset hospital. As the organisation's Endpoint Administrator, she has been assigned permissions to maintain and manage her organisations information on the Endpoint Catalogue Service (EPCSe) to ensure information is recorded and maintained to enable system to system interaction.

Jayne accesses the EPCSe using her login credentials. As she has permitted rights to manage the service, she is able to create and amend organisational endpoint templates.

Jayne's organisation has a single solution for a single connection type and single (on-prem) endpoint e.g. deploying BaRS under a self-managed endpoint on-prem.

Jayne can view the current templates and those on display do not meet her organisation needs, so she configures the endpoint details manually, including:

- Product ID
- url
- connection type

thereby establishing a 'Private Endpoint Template' which can be viewed and managed by her organisation only.

2.3.4 Scenario 4: Single Solution with Multiple Connection Types and Multiple (On-prem) Endpoints

Jamie works for Ferry Port Emergency Department, part of Croydon Hospital. As the organisation's **Endpoint Administrator**, he has been assigned permissions to maintain and manage his organisations information on the Endpoint Catalogue Service (EPCSe) to ensure information is recorded and maintained to enable system to system interaction.

Jamie accesses the EPCSe using his login credentials. As he has permitted rights to manage the service, he is able to create and amend organisational endpoint templates.

Jamie's organisation has a single solution for multiple connection types and multiple (on-prem) endpoints e.g. an ITK connection, BaRS deployed under a self-managed endpoint on-prem and email. These would be returned in a prioritised list based on preferred communication preference or sender system connection maturity.

Jamie can view the current templates and those on display that do not meet his Healthcare Service needs, so he configures the endpoint details manually, including:

- Product ID
- url
- connection type

thereby establishing a 'Private Endpoint Template' which can be viewed and managed by his organisation only.

The organisation can manage endpoint template it creates by assigning to its Healthcare Service by connection type and endpoint independently and these are exclusive to his organisation

2.3.5 Scenario 5: Multiple Solutions with Multiple Connection Types and Multiple (On-prem or Federated) Endpoints and Order of Precedence Defined

Angela works for Leeds Brooke Street Pharmacy as the organisation's **Endpoint Administrator**, she has been assigned permissions to maintain and manage her organisations information on the Endpoint Catalogue Service (EPCSe) to ensure information is recorded and maintained to enable the priority of system-to-system interaction.

Angela accesses the EPCSe using her login credentials. As she has permitted rights to manage the service, she is able to create and amend organisational endpoint templates.

Angela's pharmacy has multiple solutions with multiple connection types and multiple (on-prem or federated) endpoints. She wants to add a BaRS endpoint for blood pressure checks service to her pharmacy and she wants to set the priority to ensure BaRS referrals are prioritised over other connection types i.e. ITK or email.

Each connection type must have a ranking, even if there is only one i.e. a single connection type would be defaulted to priority status of '1'.

2.3.6 Scenario 6: Single Healthcare Service with Multiple Connection Types and Multiple Endpoints (Managed under SaaS)

Anita works for Newcastle Bridge Pharmacy as the organisation's Endpoint Administrator, she has been assigned permissions to maintain and manage her organisations information on the Endpoint Catalogue Service (EPCSe) to ensure information is recorded and maintained.

Anita accesses the EPCSe using her login credentials. As she has permitted rights to manage the service, she is able to create and amend organisational endpoint templates.

Anita's pharmacy has multiple solutions from different system suppliers which include SunRay, Pharm1, OctHealth. Each of these will have multiple connection types and multiple endpoints. When Anita is ordering her system suppliers in her view of the EPCSe, she will append her pharmacies email address to each of the supplier endpoint templates i.e. the same email to be used by all her system suppliers.

Through the EPCSe Anita can see which system supplier is supplying which Healthcare Service is offering what service. For example, she notes the following:

- SunRay is supplying solution for capturing blood pressure
- Pharm1 is supplying the solution for capturing contraception
- OctHealth is supplying the solution for capturing Pharmacy First and smoking cessation

2.3.7 Scenario 7: Single Healthcare Service with Single Connection Types and Single (On-prem) Endpoints

George works for St James GP Practice. As the organisation's Practice Manager, he has been assigned permissions to maintain and manage his organisations information on the Endpoint Catalogue Service (EPCSe) to ensure information is recorded and maintained to enable system to system interaction.

George accesses the EPCSe using his login credentials. As he has permitted rights to manage the service, he is able to create and amend organisational endpoint templates.

George wants to stand a new respiratory hub for his practice. He contacts the DoS team to advise them of his intentions. He provides the DoS team with all relevant service information required by DoS to enable them to either identify an existing DoS ID or create a new service for this GP practice that in turn would generate a new unique service ID for the respiratory hub George's practice is standing up.

George's practice has commissioned NewHealth IT system supplier to provide the solution for the new respiratory hub service. George provides NewHealth IT with the unique DoS ID that has been created for this service. NewHealth IT access the EPCSe and create a new endpoint template, including the new id along with other information they need to capture.

The new endpoint template is now available for George to view and manage as required.

3.0 Data Model

At the highest level of abstraction, the EPCSe data model can be described by the diagram in Figure 1 and Figure 2, as follows:

- A **Healthcare service** is provided by an **Organisation**. Examples include hospital clinics for elective care, emergency departments, pharmacy services, GP surgeries, non-NHS Services or private services.
- Digital retrieval and the management of the Healthcare services' data is represented by one or more **endpoints**. The endpoints can be used for system-to-system messaging to carry out a range of operations such as viewing of medical records, creating referrals, or booking, cancelling, and rescheduling appointments.

The endpoints are typically managed by the system suppliers of digital services i.e. the software manufacturer organisations. This is usually the case where the software company runs their digital services from a central location such as the public cloud. In some cases, however, the software is installed as an off-the-shelf product, hosted on-premises, and managed by the NHS organisation itself. A special case is an assured endpoint that is not linked to any healthcare service. These endpoint templates will be used when configuring a Healthcare Service based on a digital service (supplier) providing the solution used by the service. The provision of an endpoint template would further provide the following benefits:

- support for bulk updates managed by the suppliers (who own the endpoints)
- consistency in the endpoint configuration

Each data type has a set of key attributes that are explained in the next sections. For a full description, see the [Endpoint Catalogue Data Model](#).

3.1 Endpoint Relationship model across organisations

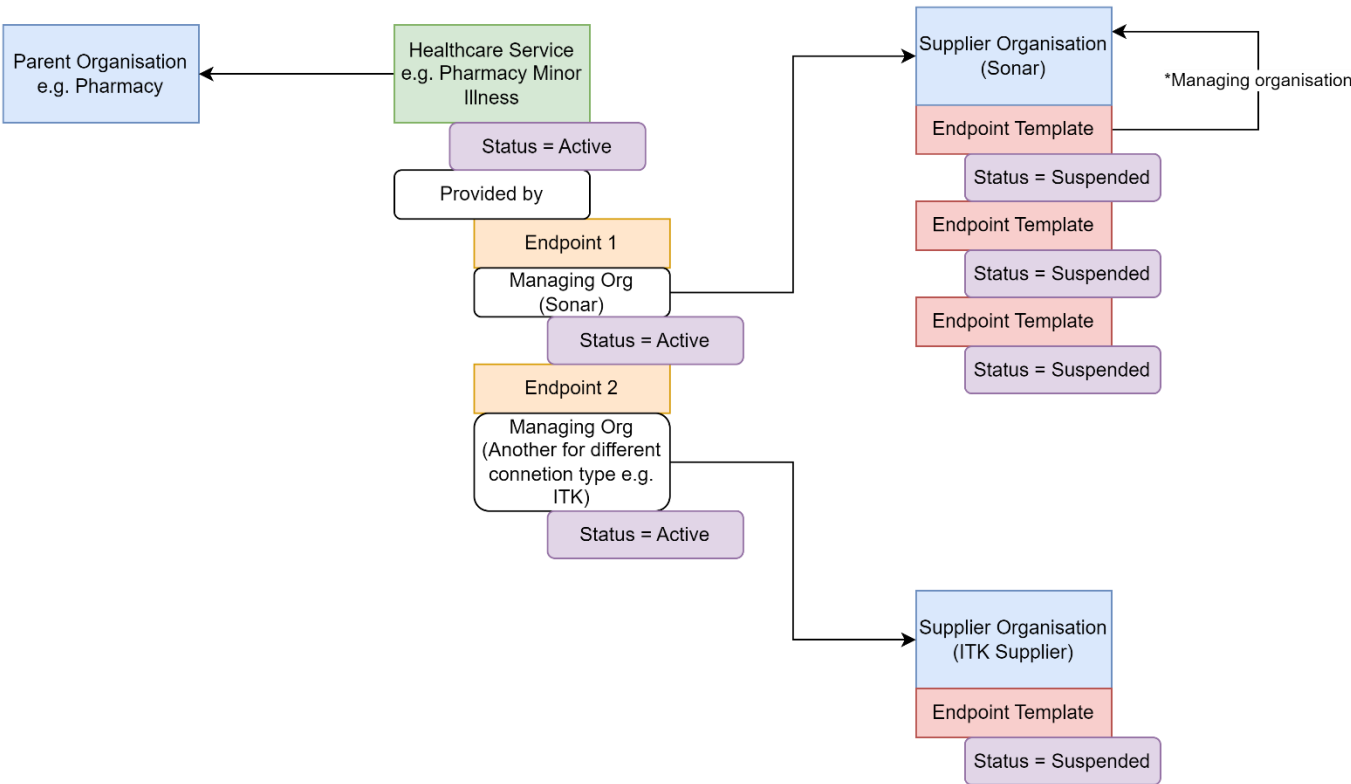


Figure 2: High-level conceptual view of an 'Active' organisation

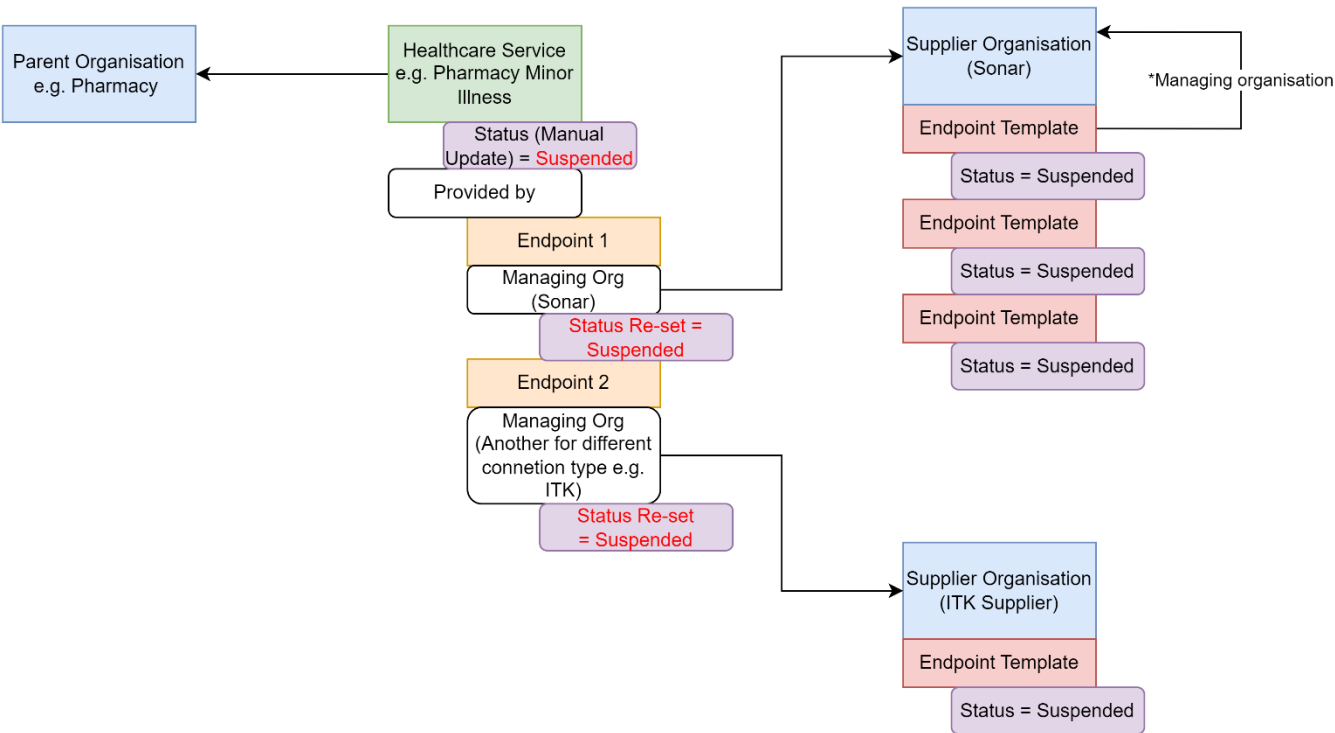


Figure 3: High-level conceptual view of an 'Suspended' organisation

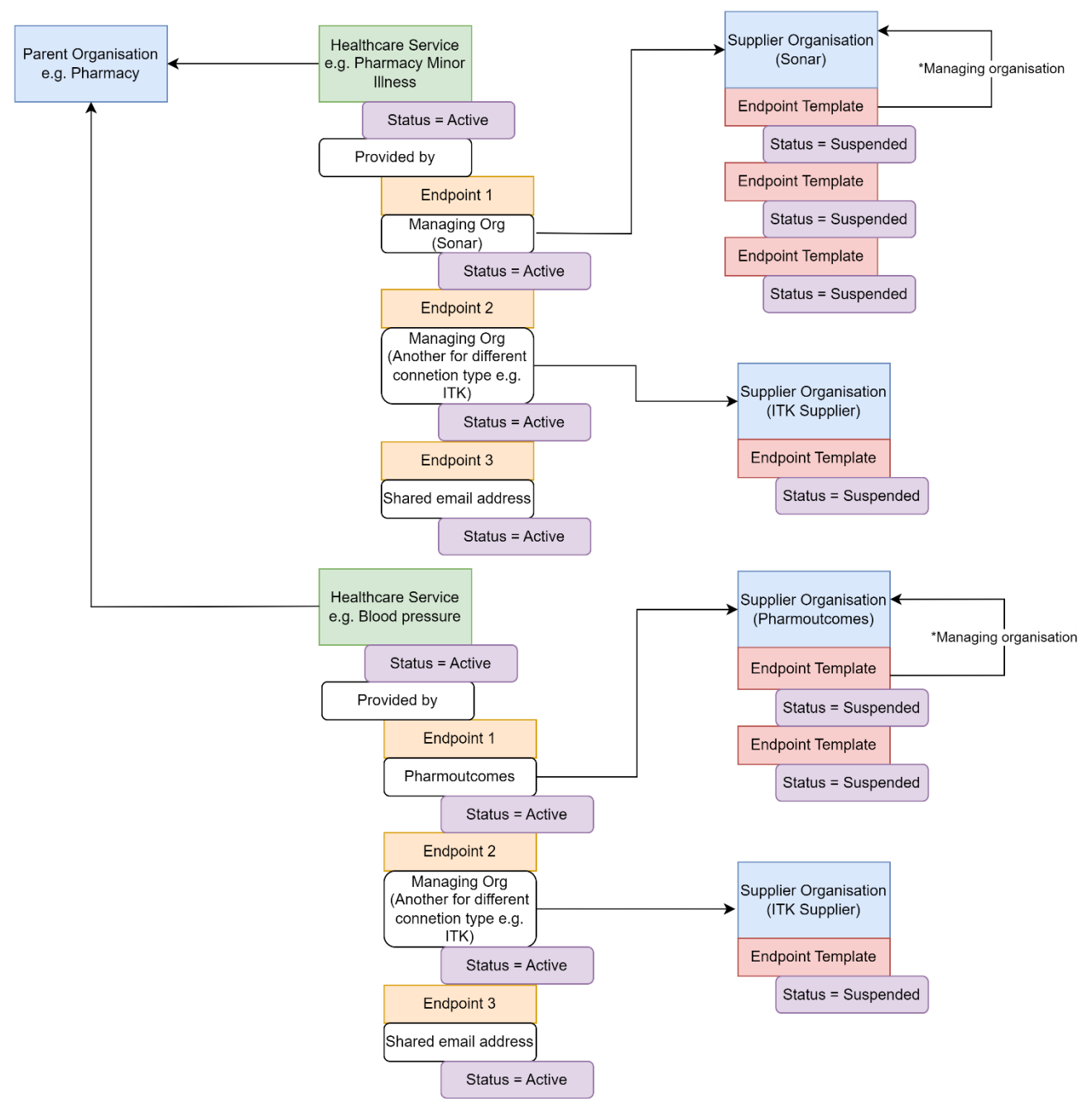


Figure 4: High-level conceptual view of single Parent Organisation with multiple Healthcare Services – All Services Active.

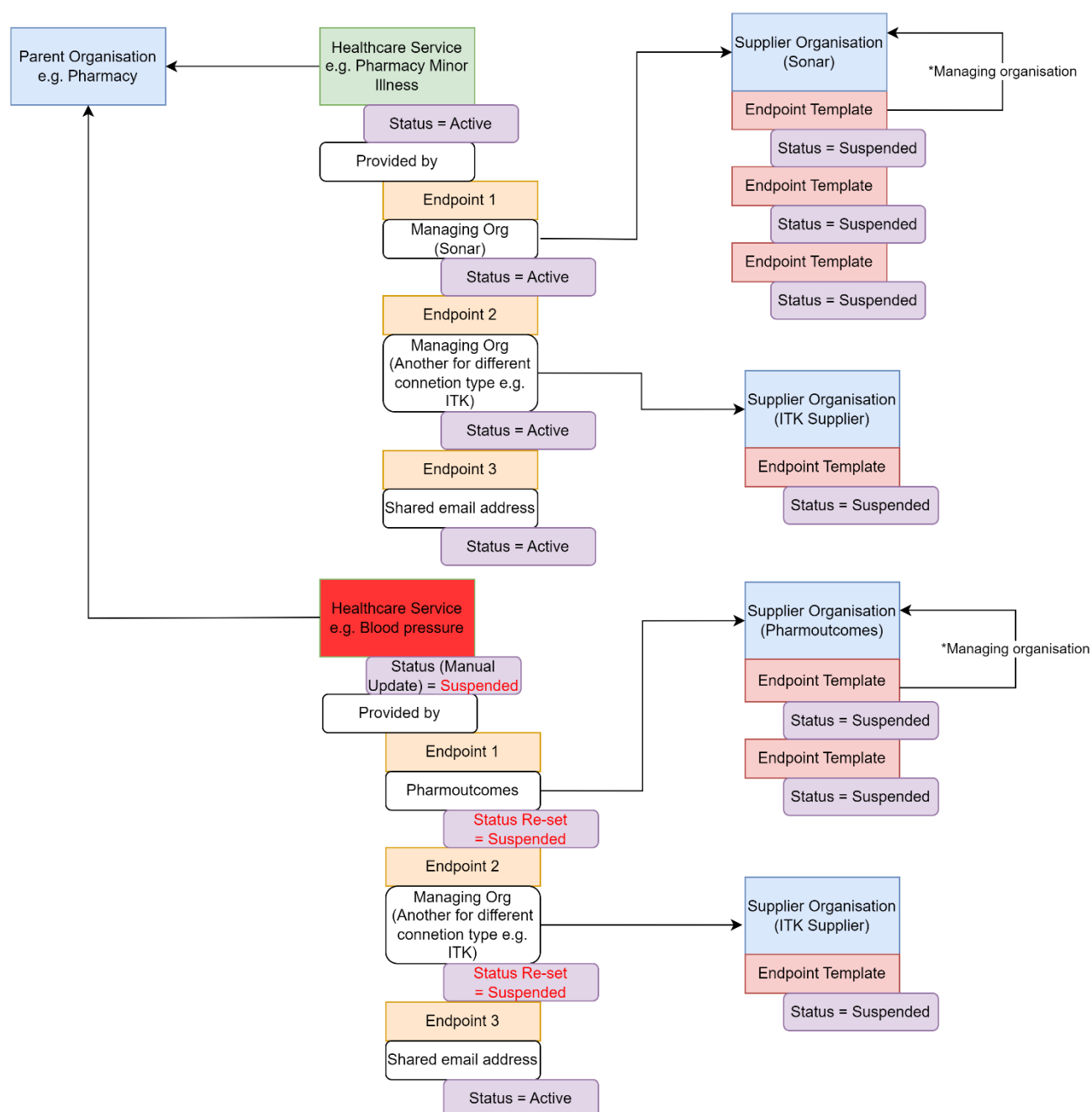


Figure 5: High-level conceptual view of single Parent Organisation with multiple Healthcare Services - One Service Suspended

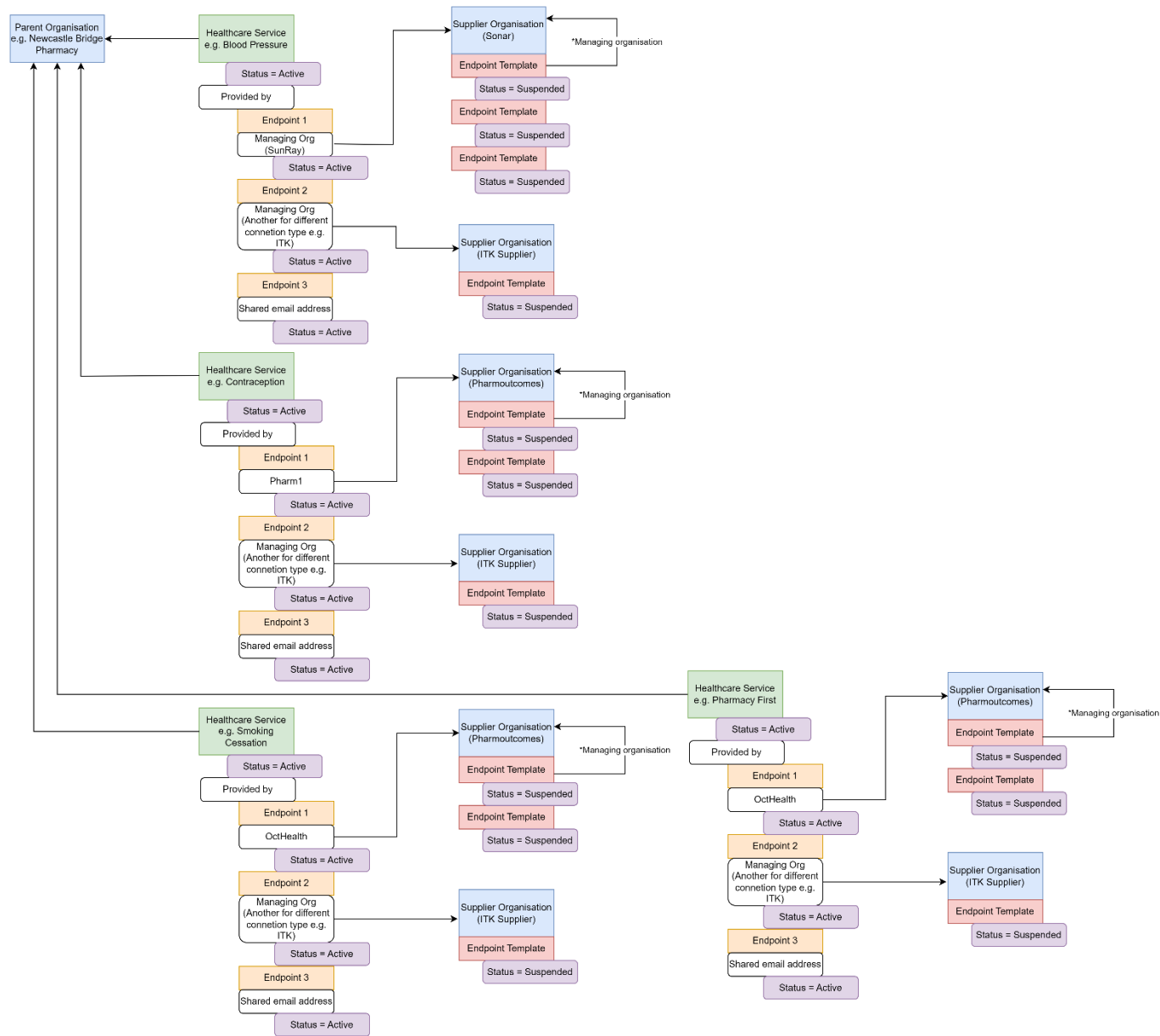


Figure 6: (See Scenario 6) High-level conceptual view of Single Health Care Service with Multiple Connection Types and Multiple Endpoints (Managed under SaaS) – All Services Active

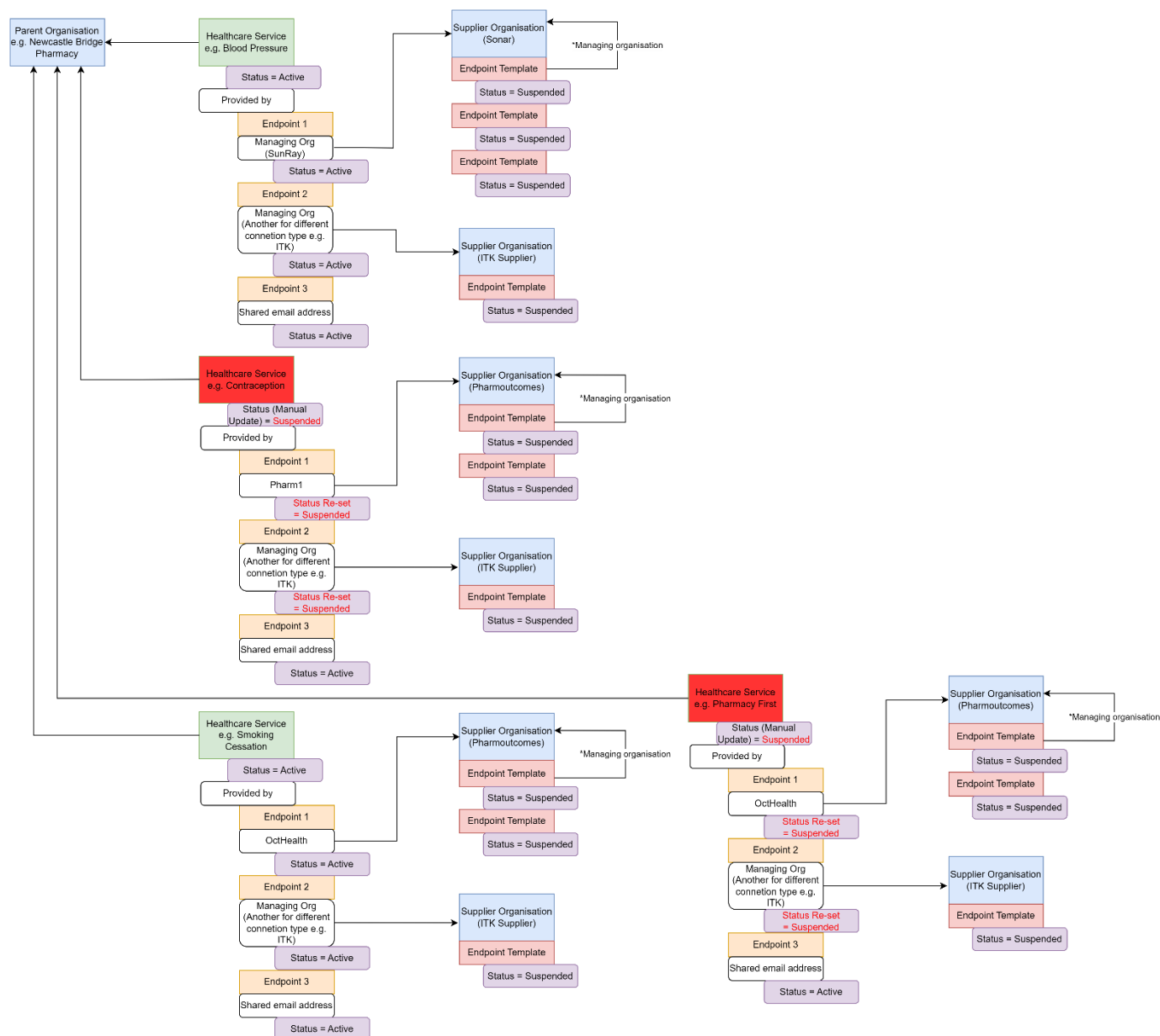
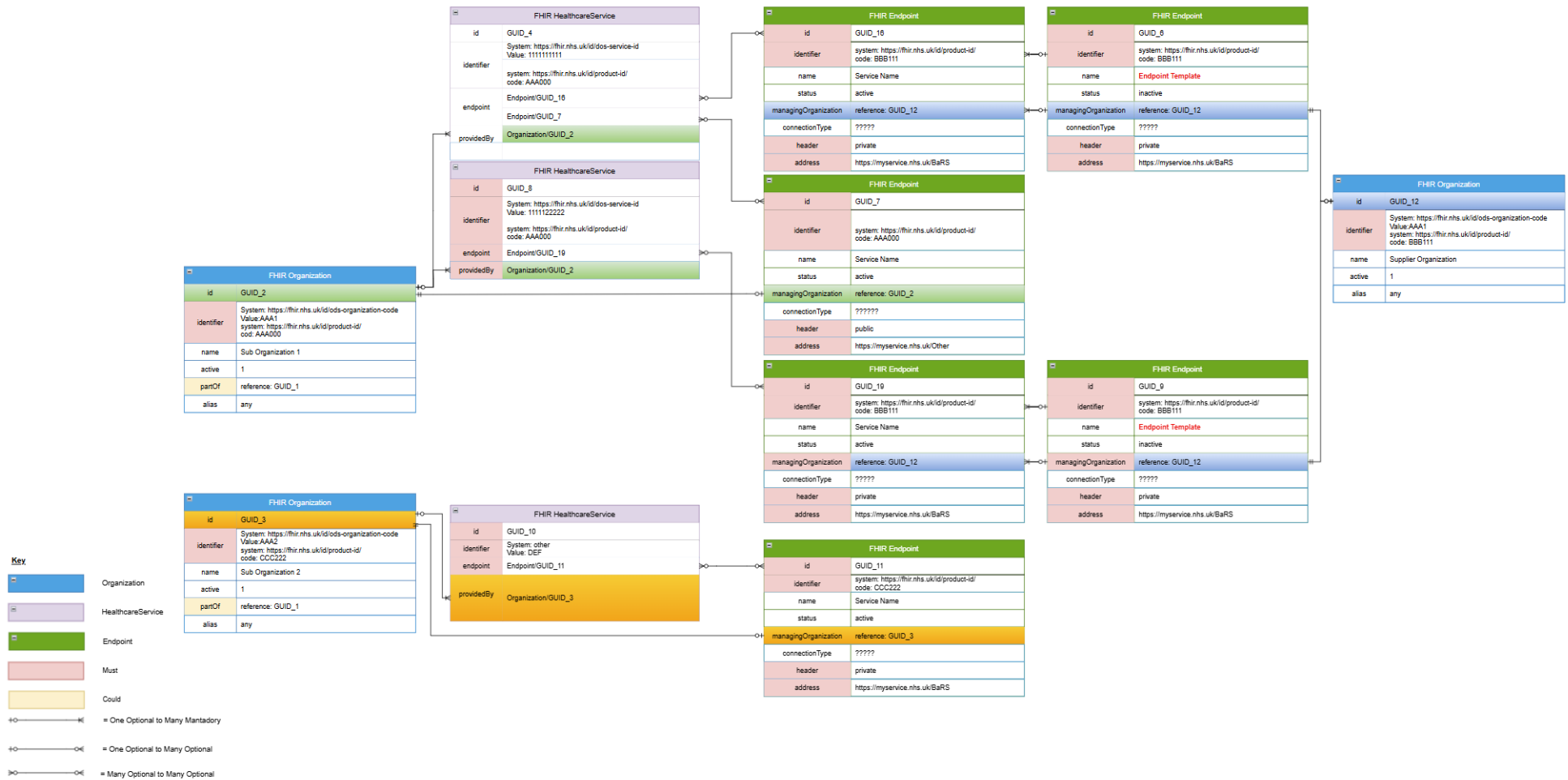


Figure 7: (See Scenario 6) High-level conceptual view of Single Health Care Service with Multiple Connection Types and Multiple Endpoints (Managed under SaaS) – Two Services Suspended

3.1.1 Supporting narrative for Figures 1 and 2 above

- **Parent Organisation** could be managing pharmacy for respective company e.g. Boots Plc = Parent Boots chemist in Leeds = child
- **Managing Organisation** in this instance would be the supplier e.g. Sonar, Pharmoutcomes, etc. The Managing Organisation could be different / unique
- ***Managing Organisation** can manage Address, Name etc.
- **Endpoint Template Status**, is this needed if the default position is 'Suspended' on creation

NOTE: This model has been provided to illustrate an architectural view of the figures above. The model is not final and will be revisited during design / build phase.



4.0 Functional Capabilities and Requirements

This section sets out the functional capabilities with respect to the endpoint template and associated service.

The capability epics within this specification have been prioritised using MoSCoW methodology.

- MUST** This word, means that a requirement is mandatory, and business and project objectives cannot be met without it.
- SHOULD** This word means that a requirement is mandatory, and information provided where available, but the system will have short-term value without it and development can therefore wait until a later increment.
- COULD** This word means that a requirement is beneficial if time or funds allow, but it is not central to project objectives.
- WILL NOT** These words indicate that a requirement will not be met in this delivery increment but may be included in the future.

4.1 Register and Manage New Endpoint (Endpoint Administrator)

	EPIC Priority (MUST)
EPCSe001	Register and manage new endpoint As an Endpoint Administrator I want to manage endpoints for my organisation So that the physical destination can be publicised to enable information sharing between service(s)
	Acceptance criterion 1: add new endpoint Given the user has access to manage endpoints When the user wants to add a new endpoint Then they shall be able to record the required information
	Acceptance criterion 2: amend endpoint Given the user has access to manage endpoints When the user wants to amend an endpoint Then they shall be able to record the required information
	Acceptance criterion 3: set status to suspended Given the user has access to manage endpoints When the user wants to set the status to suspended Then they shall be able to record the required information
	Acceptance criterion 4: set status to private Given the user has access to manage endpoints

	When the user wants to set the status to private Then they shall be able to record the required information
	Acceptance criterion 5: find endpoint template Given the user has access to manage endpoints When the user wants to find a specific endpoint template Then they shall be able to carry out search with a set of optional parameters, including but not limited to: • Connection type • Name • Product ID • Supplier ID
	Acceptance criterion 6: read endpoint information Given the user has access to manage endpoints When the user wants to read endpoint information Then they shall be able to view and read the required information regardless of its status
	Additional Implementation Details Information required for endpoint includes: • Product ID • url • Connection Type Note1 : All identifiers as set out within the Identifier section of this document Note 2: Some of the Endpoint Template will only be visible to specific users i.e. Private to that Organisation or Healthcare Service.

4.2 Set-up and Manage Endpoint Templates (Supplier)

	EPIC Priority (MUST)
EPCSe002	Set-up and manage endpoint templates (Supplier) As a Supplier I want to register an endpoint template for a federated solution So that any Healthcare Service configuring endpoints adopts the globally defined endpoint template
	Acceptance criterion 1: add new endpoint Given the user has access to manage endpoints When the user wants to add a new endpoint template Then they shall be able to record the required information
	Acceptance criterion 2: amend endpoint Given the user has access to manage endpoint template

	When the user wants to amend an endpoint template Then they shall be able to amend the required information AND The change must be cascaded to all active inherited endpoints
	Acceptance criterion 3: read endpoint information Given the user has access to manage endpoint template When the user wants to read endpoint information Then they shall be able to view and read the required information regardless of its status
	Additional Implementation Details The status of an endpoint template is fixed i.e. cannot be modified. Where a supplier BaRS implementor has defined an endpoint definition (template) for a given product, it must adopt the supplier's endpoint definition (template). Note 1: Healthcare Services configured for a product where a supplier had defined an endpoint must use the supplier's definition from their endpoint template. If supplier BaRS implementor updates their endpoint definition (template), any Healthcare Service endpoint which are based on it must inherit any change. Note 2: This change must support bulk update of endpoints Note 3: Bulk import is currently out of scope and will require further consideration to bring it into scope

4.3 Define Endpoint Without Associated Healthcare Service (Supplier)

	EPIC Priority (MUST)
EPCSe003	Define endpoints without associated Healthcare Service (Supplier) As a Supplier I want to create an endpoint template without reference to a Healthcare Service So that an Endpoint Administrator aligns the with Suppliers definition of their endpoint
	Acceptance criterion 1: add new endpoint Given the user has access to manage endpoints When the user wants to add a new endpoint Then they shall use the endpoint template that matches the connection type and product ID defined by the supplier
	Additional Implementation Details Any constraints to be defined during elaboration of the data model

4.4 Define Hierarchy for Setting Communication Preferences

	EPIC Priority (MUST)
EPCSe004	Define endpoints hierarchy As an Endpoint Administrator I want to define the priority for communicating preference So that I can set a prioritised list based on preferred communication preference or sender system connection maturity.
	Acceptance criterion 1: add communication priority Given the user has access to manage endpoints When the user wants to set the hierarchy for communication preferences Then they shall be able to set a prioritised list based on preferred communication preference or sender system connection maturity.
	Acceptance criterion 1: amend communication priority Given the user has access to manage endpoints When the user wants to amend the hierarchy for communication preferences Then they shall be able to amend the prioritised list based on preferred communication preference or sender system connection maturity.
	Additional Implementation Details - Hierarchy for fallback if messaging fails • FHIR • ITK • Email

4.5 EPCSe to Present Information in Human Readable Format

	EPIC Priority (MUST)
EPCSe005	Present information in human readable format As an Endpoint Administrator I want information within the EPCSe to be in human readable format So that I do not have to interpret, or look-up coded values returned
	Acceptance criterion 1: add new endpoint Given the user has access to manage endpoints When the user wants to view information within the EPCSe Then the shall information returned shall be in human readable format

	Additional Implementation Details Note: The display of information in a user interface if for future consideration
--	--

4.6 Provision of Audit

	EPIC Priority (MUST)
EPCSe006	Provision of Audit As an Endpoint Administrator I want a full audit history of any changes applied to an endpoint So that I have a clear view of when endpoints are added, amended, changed status, etc. for audit and support purposes
	Acceptance criterion 1: add new endpoint Given the user has access to view endpoints When the user wants to review changes to endpoint Then they shall have complete transparency of change and duration of endpoint management changes
	Additional Implementation Details The solution must provide as a minimum detail of all changes applied using the following parameters for searching: - Date - Date Range - Organisation identifier - Product Identifier - Endpoint Identifier - Status of endpoint - Healthcare Service Identifier - Product ID - url - Connection type

4.7 Reporting

	EPIC Priority (MUST)
EPCSe007	Provision of Reporting As an Endpoint Administrator I want report on activity and management of the service So that I have a clear view of when endpoints are added, amended, changed status, requested (interrogated) etc. for reporting purposes
	Acceptance criterion 1: add new endpoint Given the user has access to manage endpoints When the user wants to view reports Then they shall have access to pre-defined reports
	Additional Implementation Details The initial report must include: - Report of administration changes (add, remove, change) - Report of interrogation of the endpoint (how many requests for a service by senders) - o Include interrogation requests where there was no associated endpoint - Report by specific dates or date ranges - o Daily - o Weekly - o 4 weekly - o User defined Note: Reporting requirements to be elaborated during the development phase of the EPCSe

4.8 Identifiers

Each instance of the data type (an *object*) in the Endpoint Catalogue database will be assigned a **GUID** (globally unique identifier).

In addition to GUIDs, each object has one or more other identifiers, as explained next. The identifiers come in pairs of attributes:

- a **value** – a string of characters (typically, alphanumeric characters and hyphens)
- a **system** – defines the format and the interpretation of the value

Each system is managed by its central body tasked with the issuance of the values for the identifiers, and with ensuring their uniqueness within that system.

4.8.1 Organisation

The **Organisation** objects ([Organization resource](#)) are uniquely identified via:

- **ODS** (Organisation Data Service) **Code**
 - o all ODS Codes are unique across the entire NHS in England
 - ODS Code vary in length and composition for different organisation types:

- pharmacies have an alpha numeric string max length 5 characters (e.g. aannn (SR124) aaann (SRA98))
- GP practices have an alpha numeric string max length 6 characters (e.g. annnnn (S56789))
- NHS trusts sites vary in composition and length but are composed of alpha and numeric characters
 - see <https://digital.nhs.uk/services/organisation-data-service>
- **Product ID**
 - an identifier issued by the Digital Onboarding Service for the specific software product assured by NHS England
 - all product IDs are unique across the entire NHS in England
 - see <https://digital.nhs.uk/developer/guides-and-documentation/digital-onboarding>
 - this identifier enhances the meaning of the Organisation objects above the basic one of “a business unit with a name, address, contact information, etc.” to all of the above with the addition of “the organisation that manages a software product”

4.8.2 Endpoint

The Endpoint objects ([Endpoint resource](#)) have the following identifiers (in addition to GUIDs):

- **Product ID**
 - the same identifier used in the related Organisation object (as explained above)
 - used to link the Endpoint object with an assured software product managed by the corresponding Organisation

4.8.3 Healthcare Service

The [Healthcare Service](#) objects have the following identifiers (in addition to GUIDs):

- **Service ID**
 - a unique identifier within a specific system, such as BaRS (Booking and Referral Standard) or UEC DoS (Urgent and Emergency Care)
- **Product ID**
 - the same identifier used in the related Organisation and Endpoint objects (as explained above)
 - used to link the business / clinical service with the digital product that supports the service’s digital operations

4.9 Other Key Attributes

4.9.1 Healthcare Service

- **Endpoint** – this is an *array* of one or more references to the Endpoint objects that can be used to retrieve and manage its digital information. The Connection Type attribute of the referenced Endpoint object provides more detail regarding what data can be managed and how; for instance, the multiple endpoints can each deal with specific

ITK interactions, or the endpoint address can even represent the main contact email address for the service

4.9.2 Endpoint

- **Address** – typically a well-formed URL that can be called directly (e.g. <https://provider.nhs.uk/fhir/v4/book> for a RESTful API, or <https://trustX.provider.nhs.uk:1880/providerMHS/Nhs111Cda> for an ITK endpoint), but all other URN types are also allowed (e.g. ASID:200000001328 for the SDS domain, or info@trustx.nhs.uk for an email address)
- **Connection Type** – indicates how to interpret the protocol, profile or standard to be used with the endpoint connection. Examples include BaRS, ITK, or SDS)
 - (needs a code set define by NHSE IOPS – as an extension of <https://hl7.org/fhir/R4/valueset-endpoint-connection-type.html>)
- **Status** – indicates whether the endpoint is active or suspended for the Healthcare Service it is linked with
- **Period** – Period is composed of the **Start** and **End** date and times defining when the endpoint is available.

4.10 High-Level Requirements

4.10.1 Organisational Requirements

ID	Rating	Requirement	Description	Notes
EPC-ORG-01	Should	Self-service	The solution SHOULD aim to support self-service methodology	This may be provided via NHSE Digital Onboarding Service, or a dedicated portal (web app), or a combination of
EPC-ORG-02	Must	API First	The solution MUST provide APIs as the main way of accessing the service's data and business logic and make those APIs available for 3 rd party use	API policies and best practice - NHS England Digital
EPC-ORG-03	Must	Internet First	The solution MUST be available on the public Internet	The APIs are to be published on the NHSE API Platform The accompanying portal (web app) must be accessible from the public Internet
EPC-ORG-04	Must	Conformance to technical standards	The solution's APIs MUST conform to NHSE's API technical standards; more specifically: • Synchronous communication • REST	API policies and best practice - NHS England Digital

			FHIR UK Core, or FHIR R4 if needed	
EPC-ORG-05	Must	UI	The solution's UI MUST conform to the NHSE Content Style and Accessibility Guidelines	Design system - NHS digital service manual

4.11 Detailed Functional Requirements

Note:

- error handling is described separately in the Non-functional Requirements section.
- all functionality should be available both via the APIs as well as any NHSE provided UI

4.11.1 Creating Data Objects

This section describes how various data objects are created.

ID	Rating	Requirement	Description	Notes/Examples
EPC-FUNC-01	Must	Registering a new Endpoint without an associated Healthcare Service	The solution MUST allow its users to register a new Endpoint without linking it to a Healthcare Service. Upon registration, the user MUST provide a fully defined Endpoint (with some exceptions) together with a fully defined Organisation that manages that Endpoint. The exceptions are: - the Endpoint Period will not be set - the Status will be set to suspended 'Fully defined' object means that values are provided for all its mandatory attributes.	This functionality allows for a newly assured <i>digital</i> service to register its endpoint and its managing organisation, without linking the endpoint to any particular Healthcare Service. These endpoint templates are used for configuration purposes only and are not linked to a Healthcare Service or available for real time workflows.
EPC-FUNC-02	Must	Registering a new Healthcare Service without an associated Endpoint	The solution MUST allow its users to register a new Healthcare Service without linking it to an Endpoint. Upon registration, the user MUST provide a fully defined Healthcare Service together with a fully defined Organisation that manages that Healthcare Service. 'Fully defined' object means that values are provided for all its mandatory attributes.	This functionality allows for the creation of a newly registered (e.g. via BSA) Healthcare Service in the Endpoint Catalogue database together with the organisation providing the service, without the need to link it to one or more endpoints. This is a scenario when a new Healthcare Service has been approved to start operating but has not yet

				decided on the supplier of its digital services.
EPC-FUNC-03	Must	Linking a Healthcare Service with one or more Endpoints	The solution MUST allow its users to link an already registered Healthcare Service that has no associated endpoints, with one or more of the available, registered Endpoint templates.	During linking, a new Endpoint object will be created (for each linkage) from an endpoint template. For each such newly created object, the Status and Period attributes will be assigned values provided by the user. There would not be an instance of linking multiple Healthcare instance to one endpoint resource.
EPC-FUNC-04	Must	Registering a new Healthcare Service, one or more new Endpoints, and linking them together	The solution MUST allow its users to register a new Healthcare Service and one or more new Endpoints, together with their managing Organizations, and to link them together in one operation.	By 'new Endpoint' we mean that a newly assured digital service is being registered in the catalogue. For each such newly registered Endpoint, the endpoint template record will be automatically created. If one doesn't already exist for that connection type/supplier
EPC-FUNC-05	Must	De-duplication of data objects	The solution MUST prevent the creation of duplicate Healthcare Service and Endpoint objects.	A Healthcare Service object is regarded as a duplicate if another object already exists with the same business identifier and business identifier type and for the same Provider organisation. An Endpoint object is regarded as a duplicate if another object already linked to the Healthcare Service exists that has the same Connection Type and an overlapping period and Managed By Organisation An endpoint template record is a duplicate of another if they have the same product ID and same connection type.

4.11.2 Updating Data Objects

ID	Rating	Requirement	Description	Notes/Examples
EPC-FUNC-11	Must	Updating Organisation objects	The solution MUST allow for the existing Organisation objects to have the following attributes updated: • name of the organisation • active status • part of • alias • ODS code	If an organisation is no longer managing a specific Healthcare Service or an endpoint, then a separate operation (described below) which updates the linkage(s) should be used The 'part of' element is FHIR reference pertaining to an organisation being a subsidiary of another organisation
EPC-FUNC-12	Must	Updating the managing and providing Organisation links	The solution MUST allow for the links between either a Healthcare Service or an Endpoint object and its managing or provider Organisation to be updated.	If the new Organisation does not already exist in the database, a new Organisation object must be created and linked as the managing organisation of the specified Healthcare Service or Endpoint object. Otherwise, the existing object should be reused. Upon creation of the new link, the old Organisation object's Active attribute should be set to False only if that organisation is no longer registered as the managing or providing organisation for any other object.
EPC-FUNC-13	Must	Updating Healthcare Service objects	The solution MUST allow for the Healthcare Service's business identifier and its system to be updated.	This operation will typically be managed under standard incident management process and used by System Admin users if those values have been erroneously entered into the database.
EPC-FUNC-14	Must	Updating Endpoint objects	The solution MUST allow for the Endpoint objects to have the following attributes updated: • status • period at once or individually.	Typically, the Status will change between Active or Suspended. The Period's start date will change if the digital service is to start sooner than initially expected. The Period's end date will typically change if either the Healthcare Service is to

				become suspended or it's switching to a new supplier.
EPC-FUNC-15	Must	Updating endpoint template records	The solution MUST allow for the endpoint template records to have the following attributes updated: • product ID • name (supplied by supplier) • address (url) The solution MUST update all Endpoint object that were created from that endpoint template which can be identified by the same product ID and the same managing organisation.	

4.11.3 Deleting Data Objects

ID	Rating	Requirement	Description	Notes/Examples
EPC-FUNC-21	Must	Deletion of Organisation, Healthcare Service, and Endpoint objects	The solution MUST prevent the deletion of any Organisation, Healthcare Service, and Endpoint object	This is an MVP requirement and will be revisited in the future when the EPCSe is more mature, and its use is well known. There may be a need to support deletion in the future. Consideration should be given towards audit purposes.
EPC-FUNC-22	Must	Deactivating an Organisation	The solution MUST allow for an Organisation object to be deactivated	This operation is only available to System Admin users. It is achieved by setting its Active attribute to False. This is a powerful operation. The consequence of deactivating an Organisation (i.e. when no new Organisation is being registered in its place) is that all the Healthcare Service provided by the organisation and all the endpoints managed by the organisation will be deactivated.

EPC-FUNC-23	Must	Deactivating a Healthcare Service	The solution MUST allow for a Healthcare Service to be deactivated.	The solution SHOULD at the same time, for safety, deactivate all of the Healthcare Service's linked endpoints (but the benefits of this need to be explored further) Deactivation must not prevent outstanding processes to be resolved for example claims to the BSA.
EPC-FUNC-24	Must	Deactivating an Endpoint	The solution MUST allow for an Endpoint to be deactivated.	

4.11.4 Retrieving Endpoints

ID	Rating	Requirement	Description	Notes/Examples
EPC-FUNC-31	Must	Retrieving Endpoints for a specific HealthCare Service	The solution MUST allow its users to retrieve all Endpoints associated with the specified Healthcare Service. The returned Endpoints MUST have their Status attribute set as Active, and the date and time at the point of retrieval MUST fall within the Periods set for the returned Endpoint object	To specify a Healthcare Service before retrieving its Endpoints, the user must specify both the business identifier and its system. For example: - identifier.system = https://fhir.nhs.uk/id/dos-service-id - identifier.value = 12345678
EPC-FUNC-32	Should	Retrieving Endpoints of specific Connection Type for a specific HealthCare Service	The solution SHOULD allow its users to retrieve all Endpoints associated with the specified Healthcare Service given the specified Connection Type. The returned Endpoints MUST have their Status attribute set as Active, and the date and time at the point of retrieval MUST fall within the Periods set for the returned Endpoint object	This allows the user to e.g. only discover the BaRS compliant endpoints.
EPC-FUNC-33	Must	Retrieving Endpoints in an Agnostic Manner	The solution MUST allow its users to retrieve all Endpoints associated with a specified business identifier of a Healthcare Service. The returned Endpoints MUST have their Status attribute set as Active, and	This functionality allows for an agnostic approach to Endpoint retrieval, where the user is unsure of the exact system associated with the business identifier. For example:

			the date and time at the point of retrieval MUST fall within the Periods set for the returned Endpoint object	identifier.value = 12345678
EPC-FUNC-34	Must	Retrieving all Endpoints	The solution MUST allow its users to retrieve all Endpoints, regardless of the values of its attributes, for reporting and housekeeping purposes.	This functionality is reserved only for the System Admin role. The user MUST be allowed to perform selective queries on specific attributes and matching values e.g. Status = Suspended, or Period = Date_1 to Date_2, etc.

411.5 Retrieving Endpoint Template Records

Some endpoints are endpoint templates and not associated with a Healthcare Service. In which case its Status will be set to suspended.

ID	Rating	Requirement	Description	Notes/Examples
EPC-FUNC-41	Must	Retrieving endpoint template records	The solution MUST allow the user to retrieve a list of endpoint template records, by using zero or more of the following Endpoint attributes as query parameters: - product ID - connection type - managing organisation's ODS code - managing organisation's name - managing organisation's alias - name	
EPC-FUNC-42	Should	Retrieving endpoint template records - Matching on Query Parameters	When retrieving the endpoint template records, the solution SHOULD allow partial, full, and case-insensitive matches on all attribute values provided in the query	

4.11.6 Validation

ID	Rating	Requirement	Description	Notes/Examples
EPC-FUNC-51	Must	Data validation	The solution MUST validate all data items (attributes) for syntactical well-formedness	Where “name” is populated within the following resources: field length and content validation will be applied. Api resource: - Organisation - Healthcare Service - Endpoint Max field length = 100 chars Format = Alpha/Numeric Only
EPC-FUNC-52	Must	Endpoint connectivity	The solution MUST ensure that an endpoint is live and can be connected to when its Status is set as Active.	This requirement is for a UI integration and not specifically for the api. This is normally carried out when a new endpoint is created, or when an existing suspended endpoint is being switched to active.

4.12 Non-functional Requirements

ID	Rating	Requirement	Description	Notes/Examples
EPC-NF-01	Must	Error handling	The solution MUST implement an error handling mechanism that is closely aligned to BaRS error handling.	https://simplifier.net/guide/nhsbookingandreferralstandard/Home/Core/1.2.0/Error-Handling/Failure-Scenarios-1.1.x?version=1.8.0 This is yet to be elaborated on (separately in Appendix 1 - Error Handling)
EPC-NF-02	Must	API Response Times	All APIs provided by the solution MUST achieve response time of 100 milliseconds, at P90.	This is taken from the live stats of the current solution for the BaRS endpoint catalogue.
EPC-NF-03	Must	Data integrity	The integrity of all data in the solution’s database MUST be preserved.	

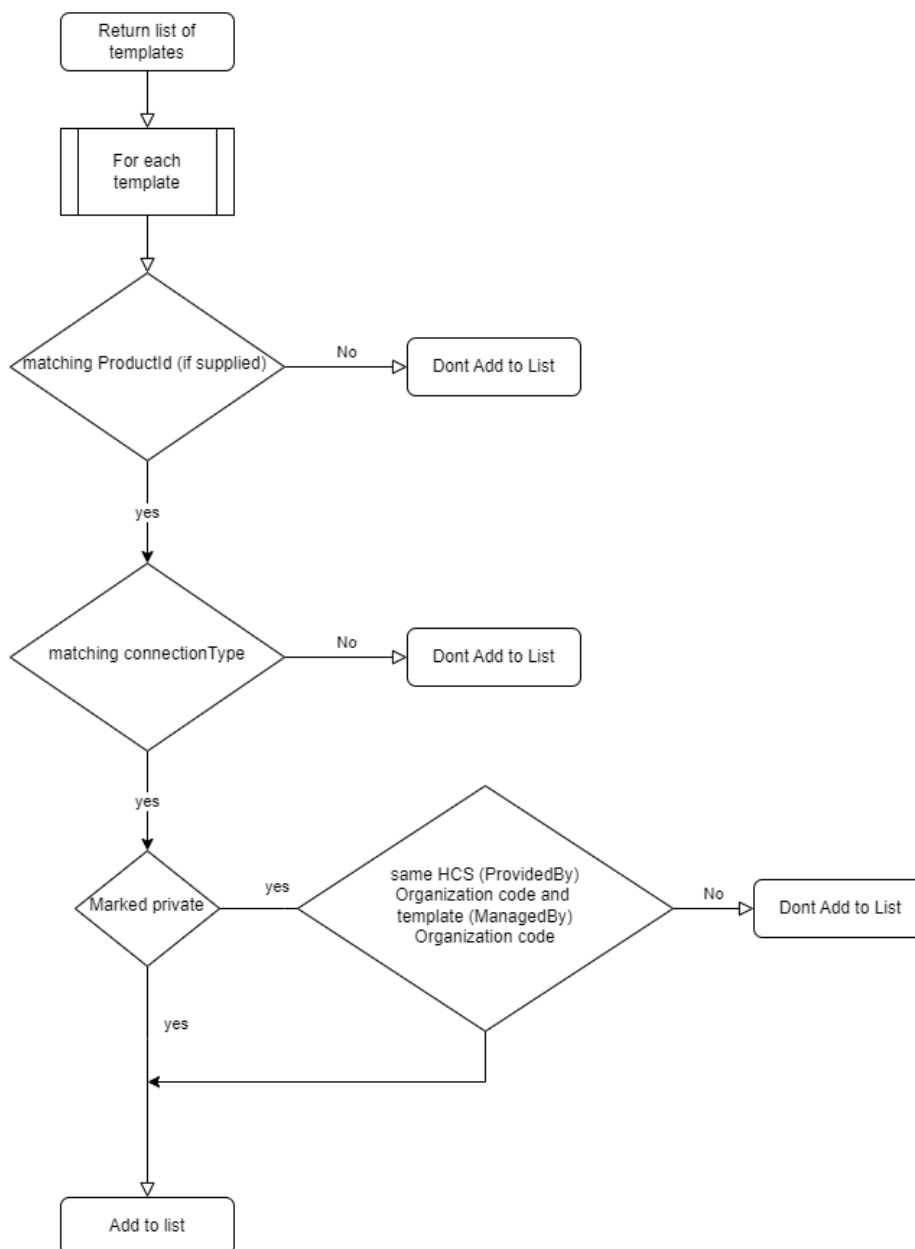
EPC-NF-04	Must	Data security	All data at rest and data in transit MUST be encrypted.	
EPC-NF-05	Must	Accessibility	The solution MUST ensure human-readable names for all Organisations, Healthcare Services, and Endpoints data objects	
EPC-NF-06	Must	Auditing	The solution MUST audit all information access attempts and actions, retaining all pertinent data therein, as well as forwarding relevant access logs to CSOC.	
EPC-NF-07	Must	Logging	The solution MUST implement operational logging to support incident troubleshooting and forward relevant log events to ITOC.	
EPC-NF-08	Should	Log events	All operational log events for the business logic SHOULD be intentional and conform to the design that has been agreed prior to the implementation of the business logic code.	The design needs to address the formatting of individual log event types, storage, retention period, and access to all relevant stakeholder groups.
EPC-NF-09	Must	API Service Class	The APIs provided by the solution MUST have the NHSE's Platinum service characteristics	See https://digital.nhs.uk/services/cloud-centre-of-excellence/cloud-security-good-practice-guide/9.-appendix-b-service-classifications
EPC-NF-10	Must	UI Service Class	The UI provided by the solution MUST have the NHSE's Silver service characteristics at a minimum	This requirement is for a UI integration and not specifically for the api. See https://digital.nhs.uk/services/cloud-centre-of-excellence/cloud-security-good-practice-guide/9.-appendix-b-service-classifications
EPC-NF-11	Must	BC/DR	The solution MUST be able to recover from a disaster scenario with no loss of data and within 2 hours from the start of the incident	RPO = 0 bytes RTO = 2 hours
EPC-NF-12	Must	Scalability	The solution MUST be dynamically scalable to allow for new consumers to	

			onboard the EPCSe without the need to manually uplift the infrastructure (e.g. via a new release)	
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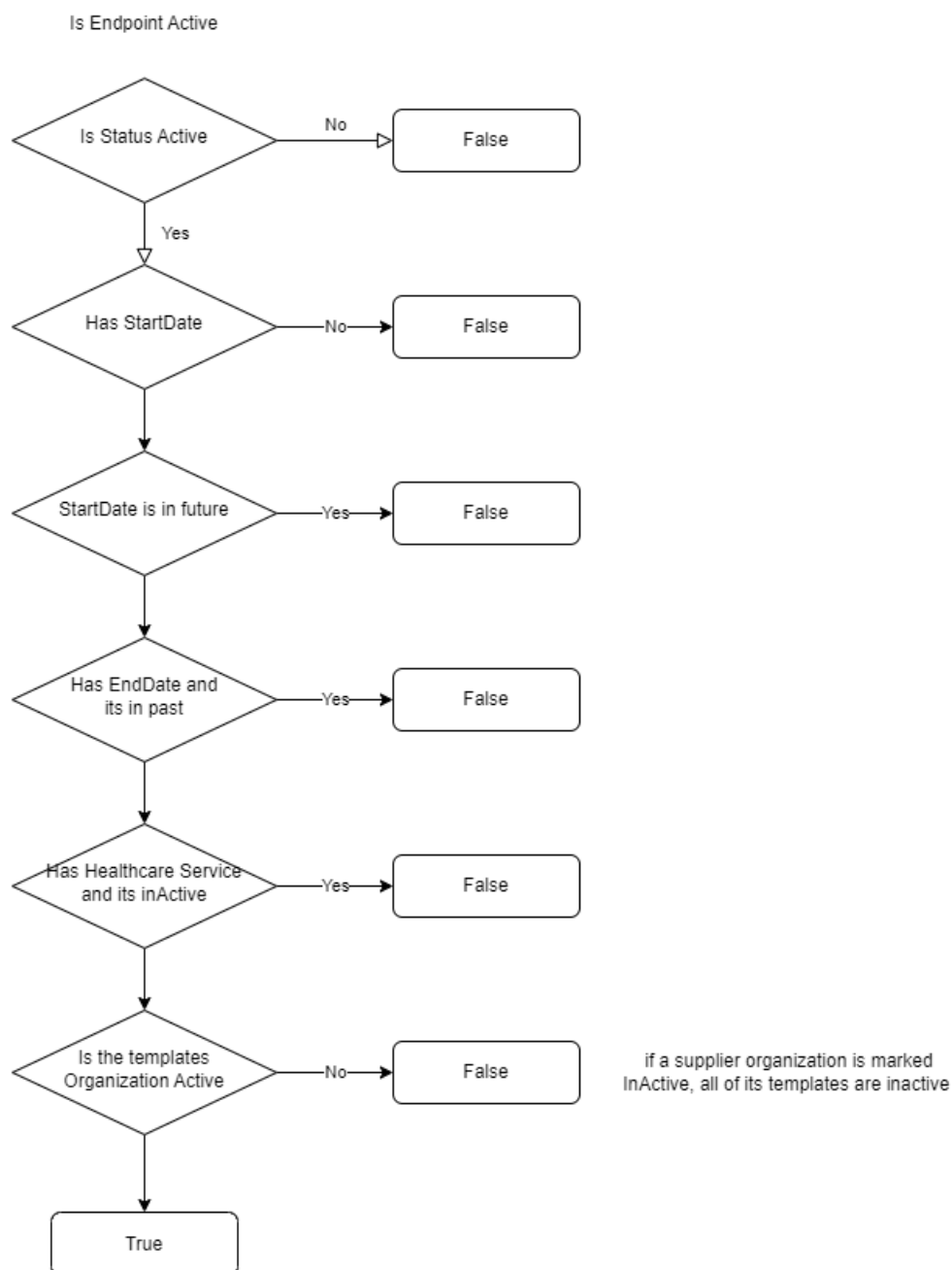
4.13 Additional Technical Requirements

Flow for getting a list of templates to present to the use for configuration.

Getting the list of templates available for a HCS to use to configure an endpoint.



Flow for if an endpoint configuration is active



5.0 RBAC Model

At the highest level, there are five distinct roles to carry out the retrieval and the management of data in the EPCSe. The access is discussed in the context of standard **CRUD** (**C**reate, **R**ead and search, **U**ppdate, and **D**ele) database operations.

The roles, the data objects, and the operations the roles can carry out on those objects are described in the Table 1 below.

NOTE: RBAC model may need further, more detailed thinking and definition during the discovery phase. The key concern and risk to address are data provenance and the protection against unauthorised modification of data. High level overview is described below, however.

Table 1. RBAC Model

Role	Objects	Access
Healthcare Service Admin	Healthcare Service and the Organisation managing the Healthcare Service	CRUD
Endpoint Admin	Endpoint and the Organisation managing the endpoint	CRUD
Activation Admin	Creating and deleting the links between Healthcare Services and Endpoints	CRUD
Consumer	Retrieval of Endpoint information when providing a business identifier.	R
“Programme” Admin	Management of all Endpoint objects for a specific type of endpoint.	CRUD
System Admin	Management of all data objects	CRUD

The authentication and authorisation processes are based on the [application-restricted access](#) and the NHS [Care Identity Service](#) and its [policies](#).

The use of the identifier “Product ID” will be used to facilitate authorisation upon a given data object. Should the client application’s assigned Product ID not match that of one within the data object, **U**ppdate and **D**ele operations will be restricted. This will ensure that data can only be modified by the owners of that data. Individual user access will be the responsibility of the client application.

This document assumes that maintenance of Endpoint information will be the same for BaRS as for UEC DoS and GP Connect.

Appendix 1 - Error Handling

Error handling will be an extension of the BaRS Core error handling design, utilising a code set within FHIR UK Core and FHIR Operation Outcomes.

“Error codes (HTTP Response codes) can be caused by any of the three parties, Client, Catalogue or Receiver involved in a transaction using the Booking and Referral API. Error codes will be accompanied by a OperationOutcome FHIR resource where possible. Error codes and their relative operation outcome codes are meant for developer use only - they should not be presented to end-users. Instead, sending applications should interpret the two codes and provide user-friendly resolution steps on failure. Sender and receiver systems should record the detailed error codes in local logs in order to support service incident investigation. There are scenarios where an operation outcome may not be applicable or available. A HTTP Response code should always be returned.

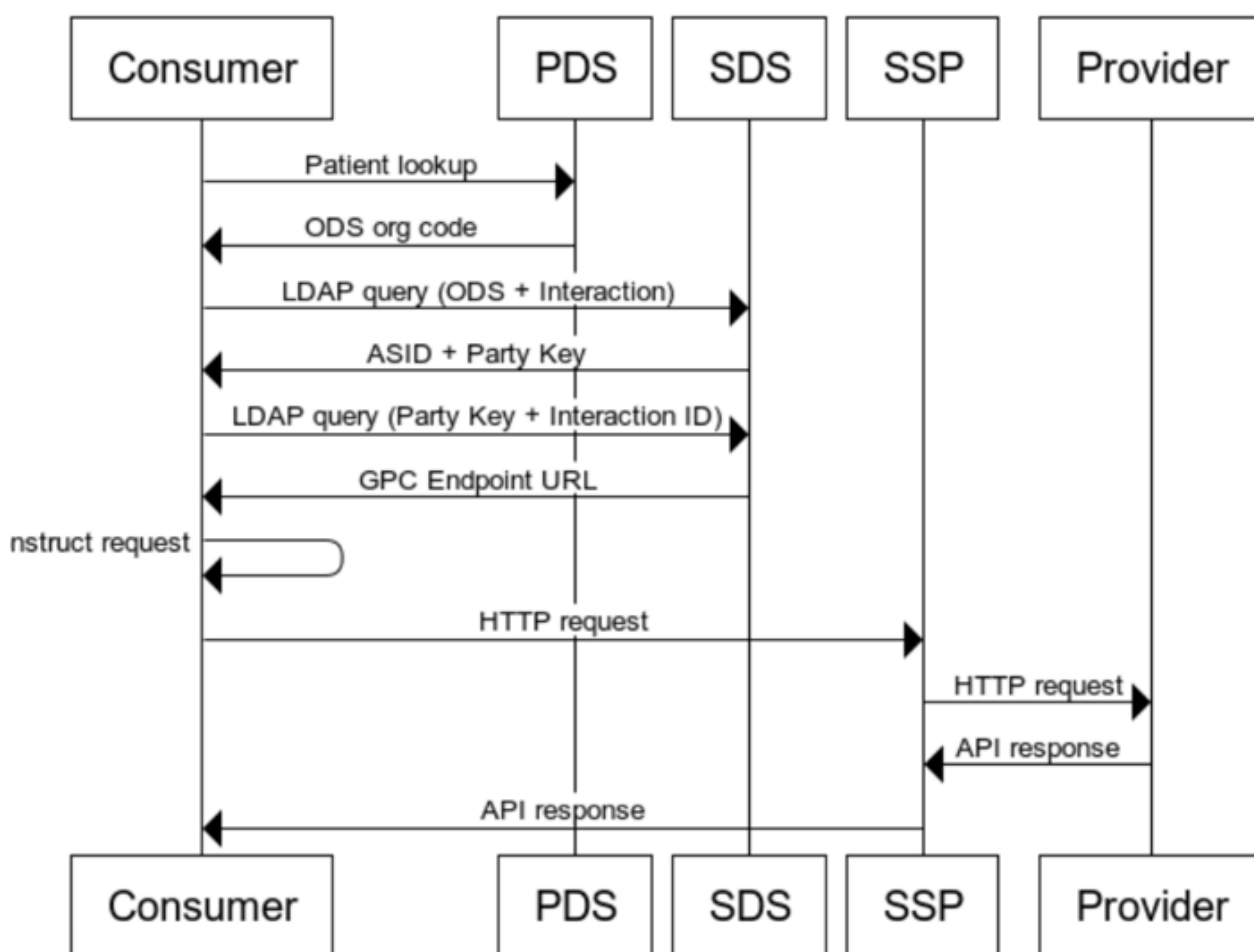
Operation Outcome codes are prefixed with an indicator of the source of the error to help identify where a problem may be.”

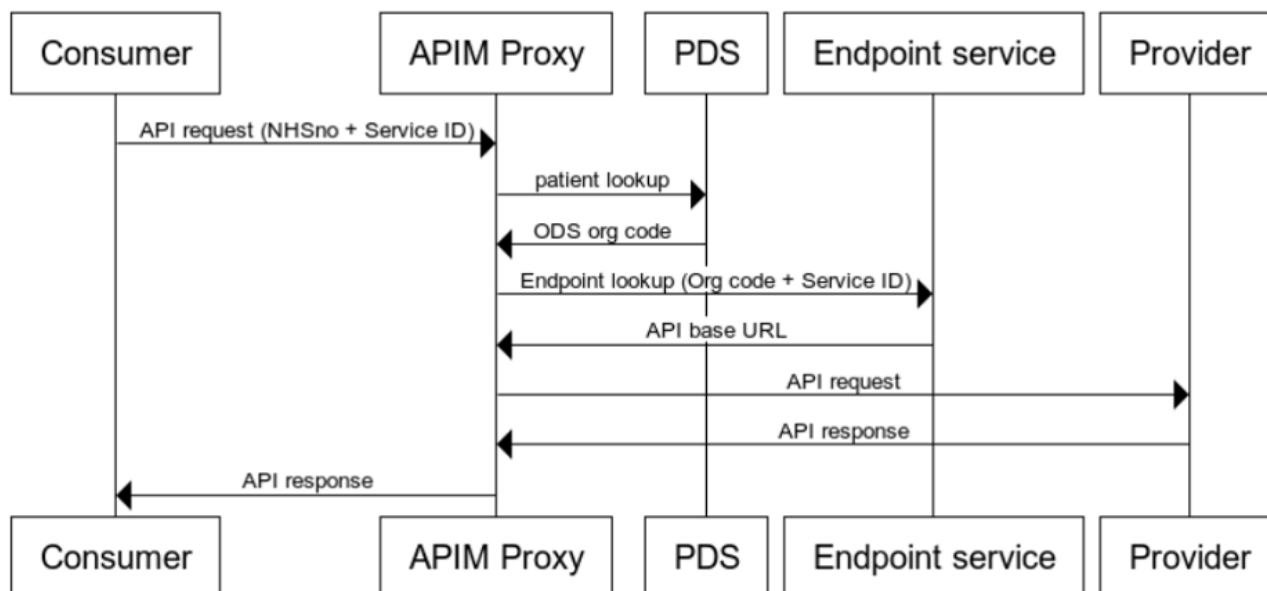
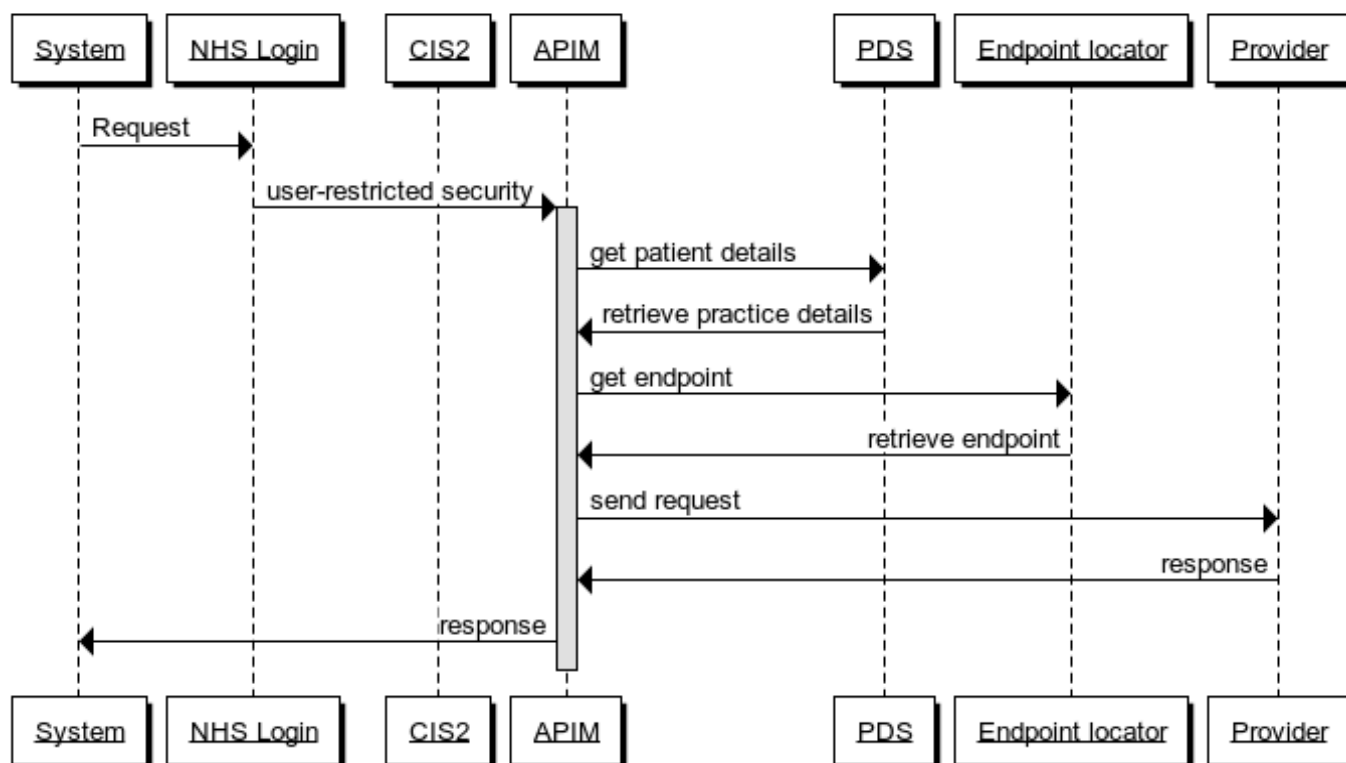
Full details of Error handling can be found on [simplifier](#). The particular scenarios which will need to be extended/adapted can be also be found on [simplifier](#).

Appendix 2 – GP Connect Supporting information

GP Connect products use either the MESH endpoint service (for Update Record and Send Document) or SDS (for Access Record: Structured, Access Record: HTML and Appointments Management). Both MESH & SDS are managed by separate BAU teams, and therefore the business processes around maintaining their endpoint catalogues are owned by those teams rather than DCAPI team. The majority of this process is now self-service, information pertaining to this can be found [here](#).

GP Connect Endpoint Lookup (as-is)



GP connect endpoint lookup (to be)**GPC New API**

www.websequencediagrams.com

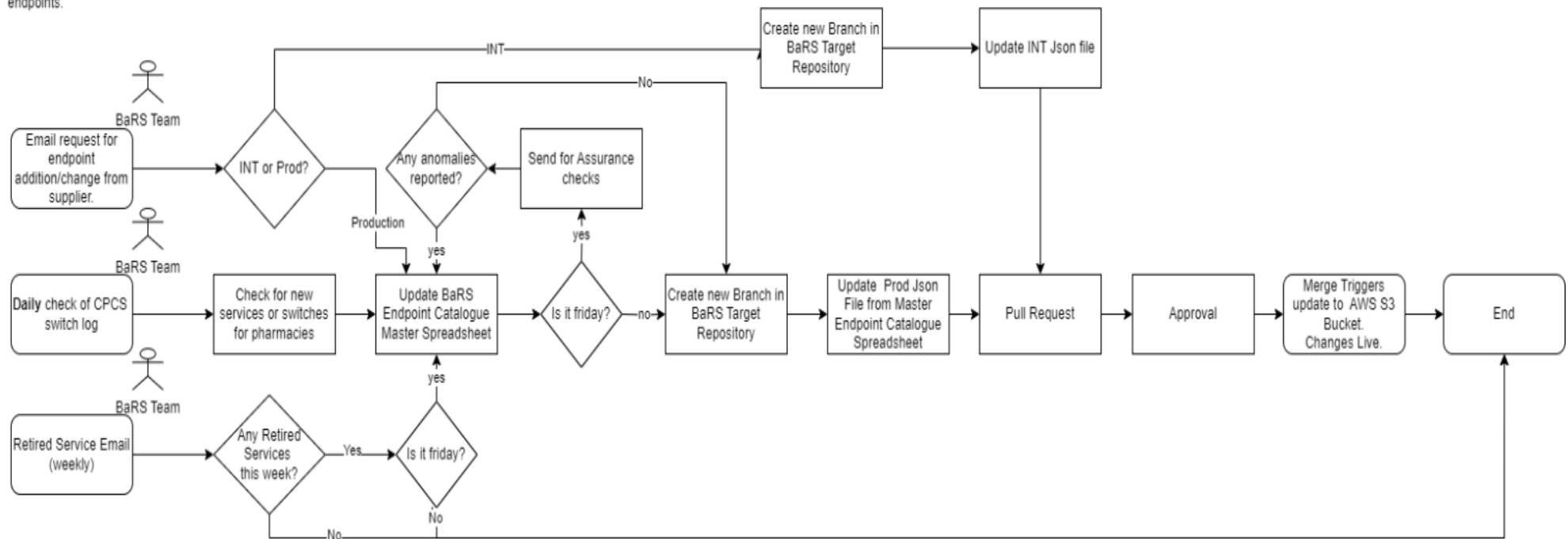
Appendix 3 – BaRS Supporting information

Business Process As-Is

BaRS Current Endpoint Business Process

The manual process for the BaRS team is detailed below, each step is performed by a member of the team.

In the case of CPCS (GP to Pharmacy) the endpoints are multitenanted and the supplier name implies the endpoint in the change. This is based off of existing information for that suppliers endpoints.

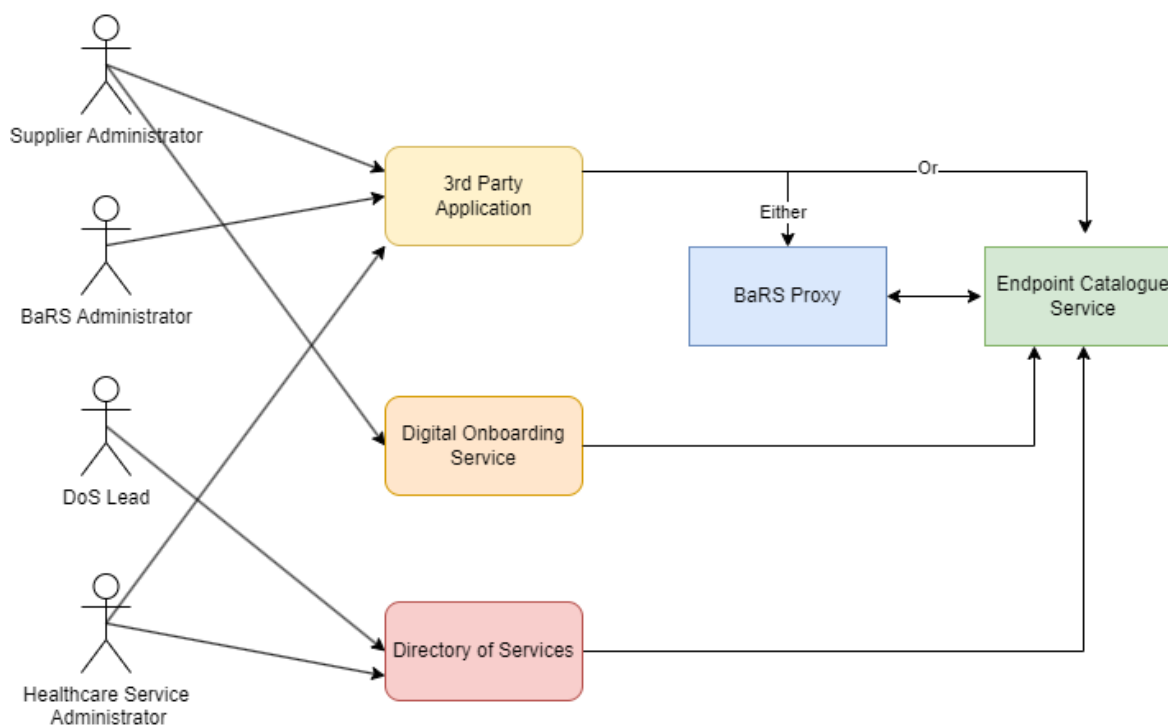


Business Process Decentralised

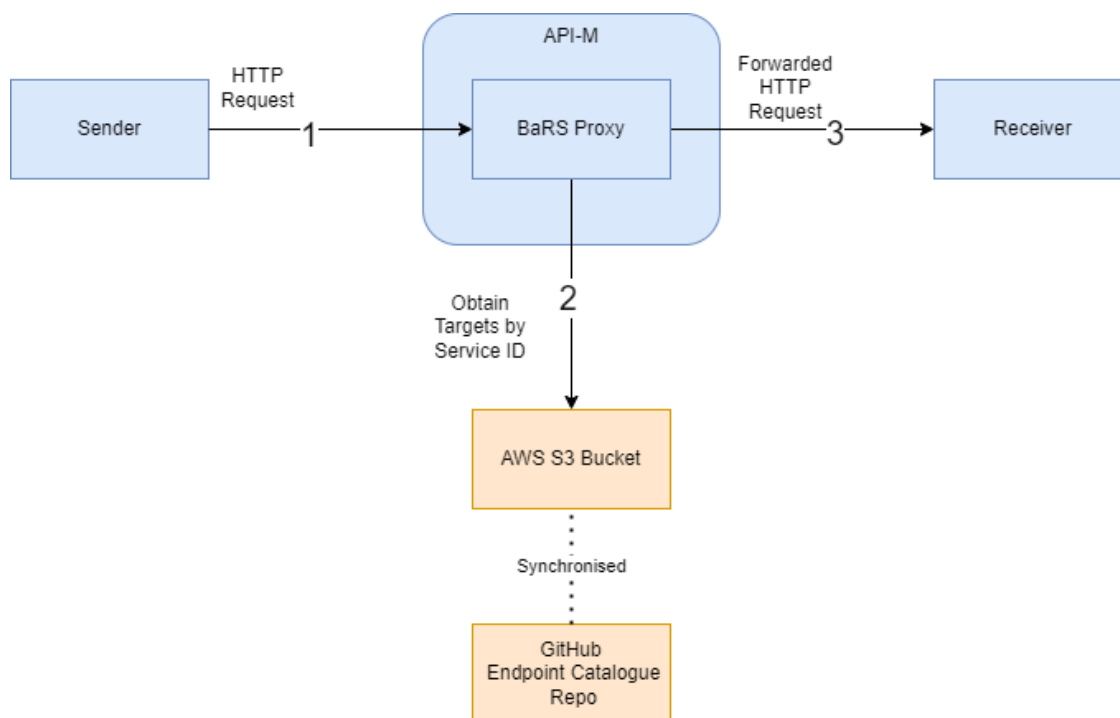
For BaRS the aim is to expose the Endpoint catalogue endpoints and act as a facade as we do with the NRL API currently, providing a SPA.

This following diagram depicts the user Use Case diagram and system behaviour specific.

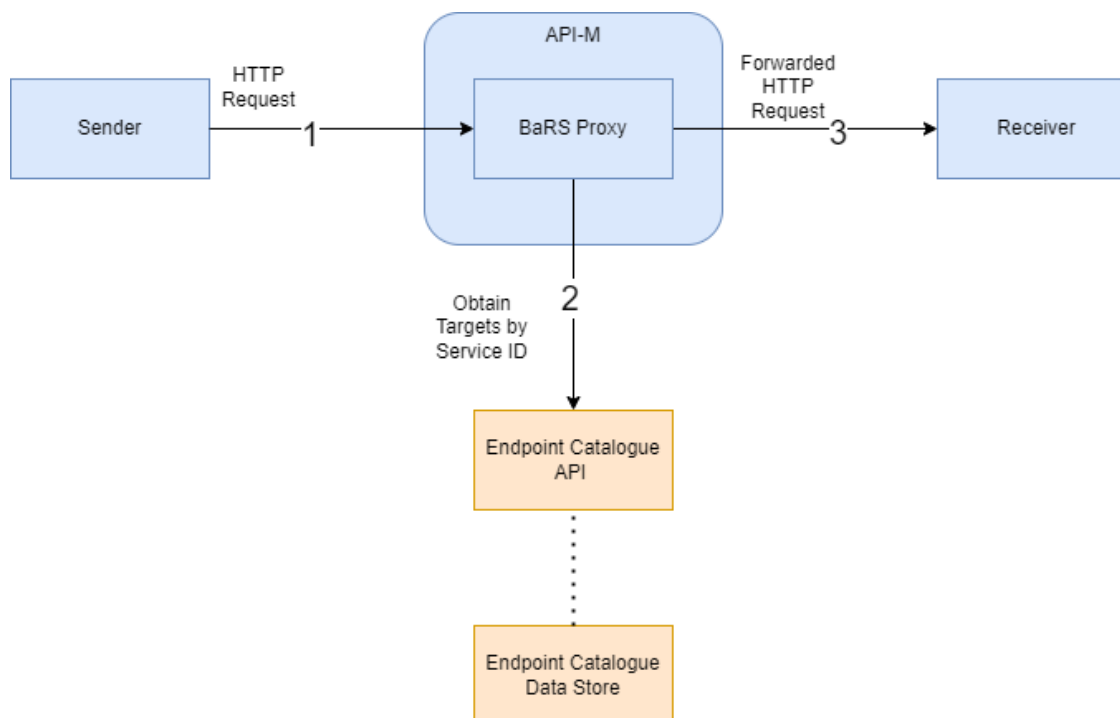
BaRS Managing Endpoints



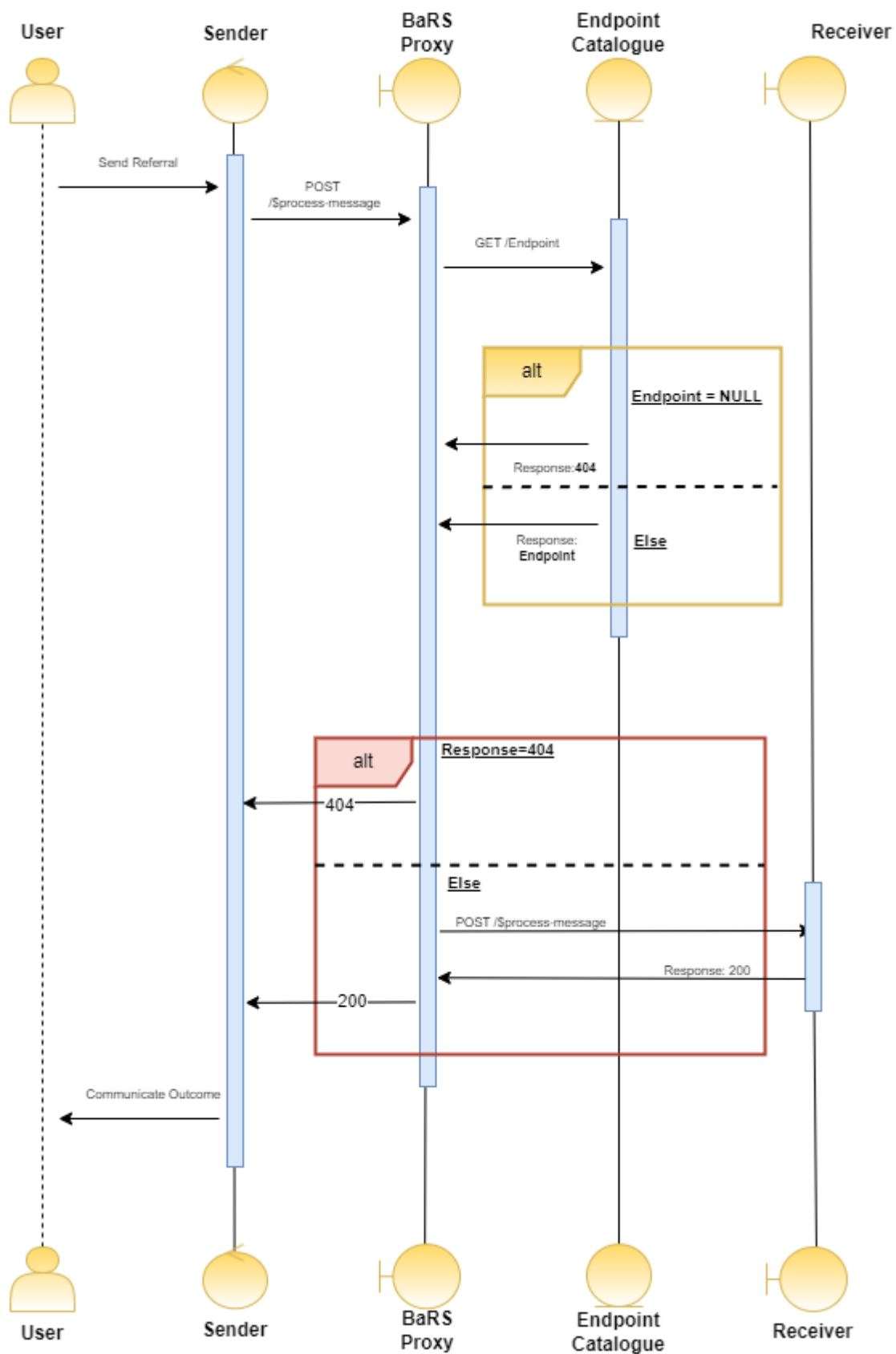
Consumer As-Is



Consumer To-Be



Routing Sequence - In this example Endpoint Catalogue represents the whole solution.

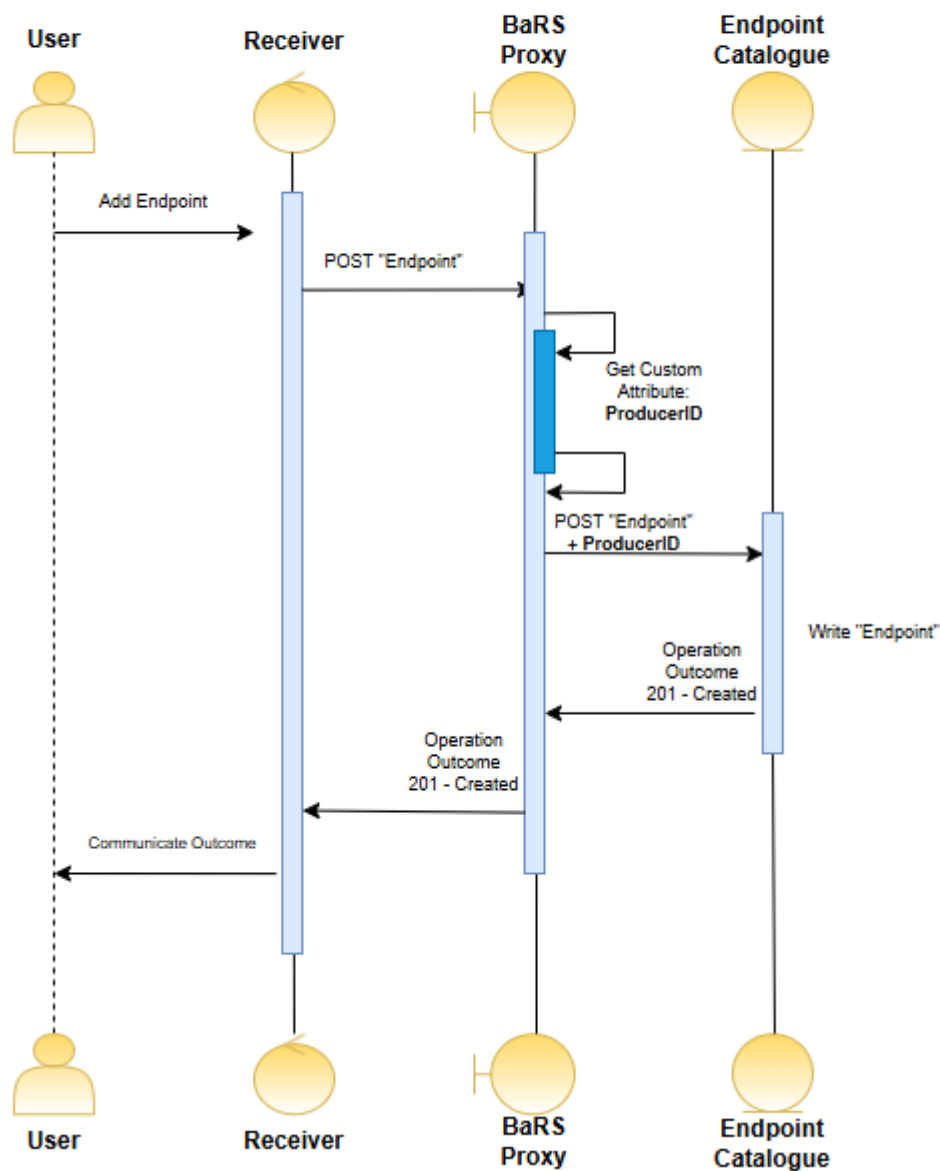


Create Sequence - In this example Endpoint Catalogue represents the whole solution.

Create

The term "Endpoint" encompasses the constituent resources that make up the BaRS concept of "Endpoint"

(Endpoint, HealthcareService, Organization)



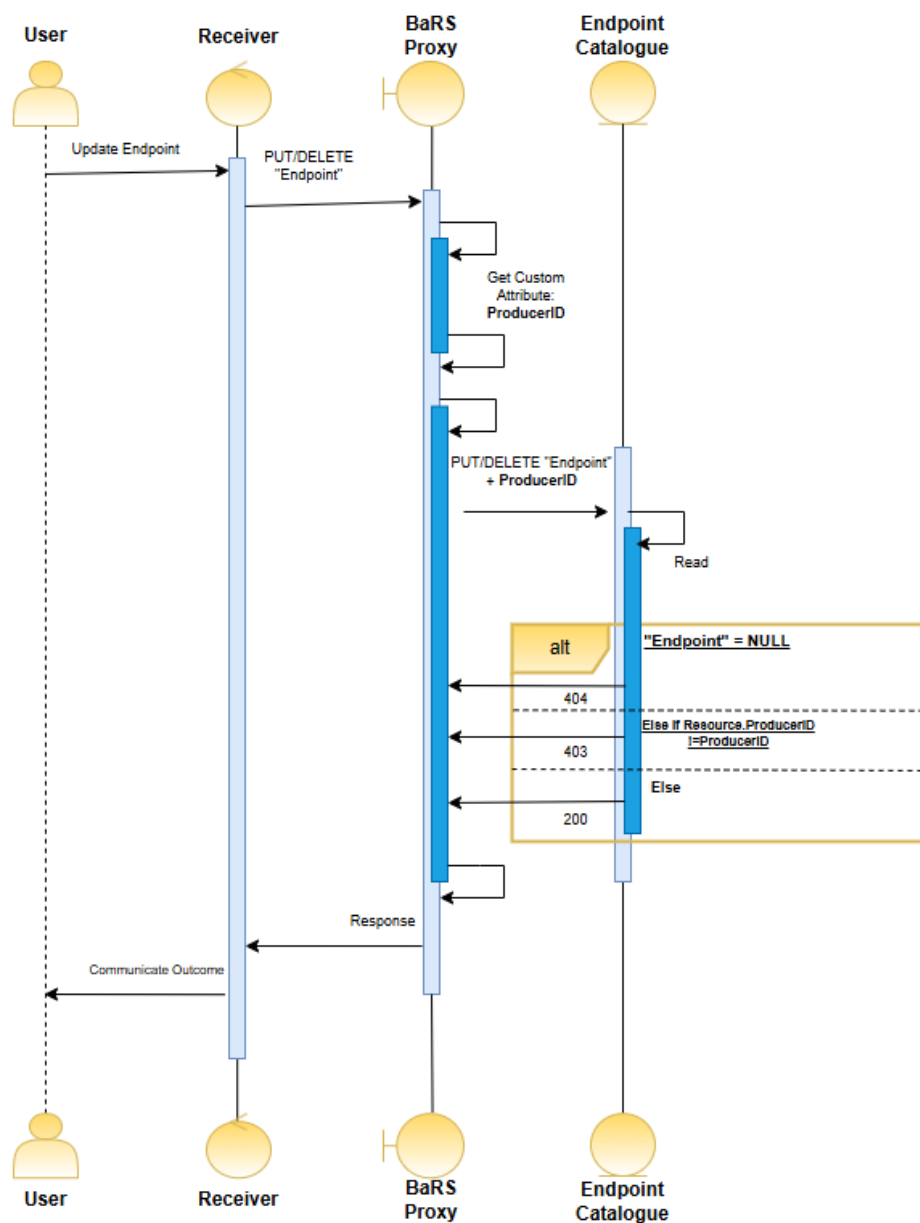
Update Sequence

Update/Delete

The term "Endpoint" encompasses the constituent resources that make up the BaRS concept of "Endpoint"

(Endpoint, HealthcareService, Organization)

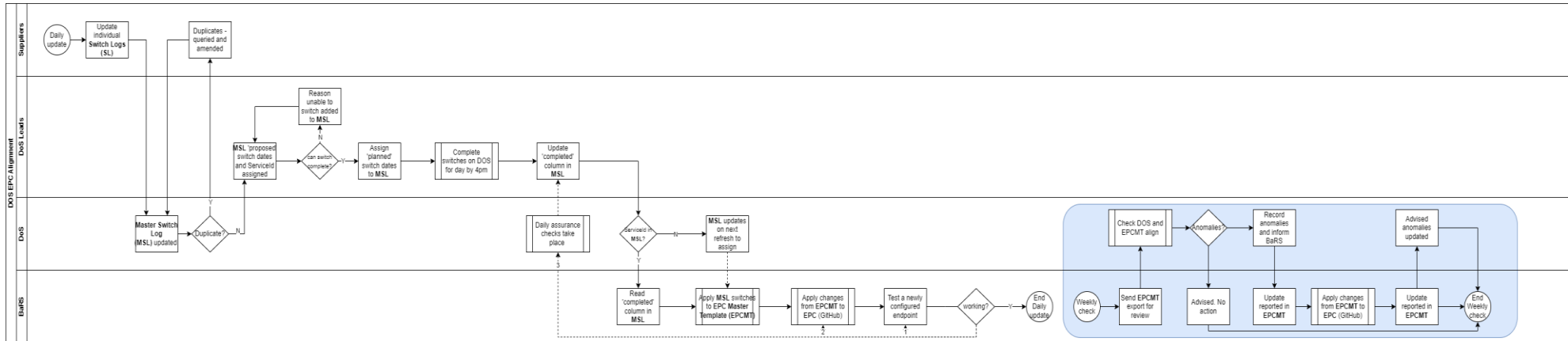
The Receiver will have to perform a read first, this is implied.



[Full proposed design on Confluence.](#)

Appendix 4 – UEC Supporting information

This is an 'end to end' process flow diagram of the UEC DoS switching process and End Point Catalogue alignment



End-to-End Process Steps:

DoS Switch Management

Activity ID	Activity Description	Team performing	Sequence	Timeframe
1	System Supplier adds pharmacy to their individual switch log by 1pm Monday , Wednesday & Friday	System Suppliers	1	Mon, Wed and Fri
2	Individual logs update Master Switch Log with checks for duplicate submissions by different suppliers	NECS DoS	2	Mon, Wed and Fri
3	The switch log is updated with switches on a Monday Wednesday Friday by 4pm.	NECS DoS	3	Mon, Wed and Fri
4	Master log updates with proposed switch date to be completed by and service IDs if available	DoS Leads	4	Daily before 4pm

5	DoS Teams review the log this can be on any given day.	DoS Leads	5	Daily before 4pm
6	DoS Teams will add PLANNED date into column J (this will be at least 1 day AFTER being added to the log) note – although this is the date it is due to be live it is not a guarantee.	DoS Leads	6	Daily before 4pm
7	If the switch cannot be completed the reason will be added to Column L and no planned date will be added.	DoS Leads	7	Daily before 4pm
8	DoS teams will complete switches on the planned date (date not guaranteed) before 4PM.	DoS Leads	8	Daily before 4pm
9	Once the switch is completed the date of completion is added to Column M	DoS Leads	9	Daily before 4pm
10	Any switches with no Service IDs will have Service IDs on the next Master Log update	NECS DoS	10	Daily before 4pm
11	Daily assurance checks also take place to support accuracy and challenge any discrepancies	NECS DoS	11	Daily before 4pm

EPC Alignment to DoS - Daily

Activity ID	Activity Description	Team performing	Sequence	Timeframe
12	Open the Master CPCS Supplier Template (MST) submissions log post 4pm daily	BaRS	1	Daily after 4pm
13	Filter the MST by "Completed Switch Date", or the date range required, for Column M	BaRS	2	Daily after 4pm
14	Open the BaRS Endpoint Catalogue Master Template (EPCMT) .	BaRS	3	Daily after 4pm
15	Turn on Conditional Formatting for duplicate cells in columns F and H.	BaRS	4	Daily after 4pm
16	Add the ODS codes from the filtered list in MST to the bottom of the EPCMT in Column F.	BaRS	5	Daily after 4pm
17	Do the same with the DoS ID (column I in MST) into column H of EPCMT .	BaRS	6	Daily after 4pm
18	If DoS ID is not present, obtain from DoS Daily extract	BaRS	7	Daily after 4pm
19	If they go red (EPCMT), this is a switch.	BaRS	8	Daily after 4pm
20	if this is a switch, find the existing entry and amend the endpoint and supplier organisation derived from existing information, in MST .	BaRS	9	Daily after 4pm
21	If they do not, this is an addition for EPCMT .	BaRS	10	Daily after 4pm

22	If this is an addition EPCMT , paste the name into column G and use the autofill from the previous entries in columns C to E, I , J and L	BaRS	11	Daily after 4pm
23	Fill in the remaining rows derived from existing information.	BaRS	12	Daily after 4pm
24	Ensure all fields are filled in the EPCMT .	BaRS	13	Daily after 4pm
25	Order the EPCMT spreadsheet by column J.	BaRS	14	Daily after 4pm
26	Create a new branch in the BaRS Targets repository.	BaRS	15	Daily after 4pm
27	Modify the live/targets.json file and update the list of endpoints.	BaRS	16	Daily after 4pm
28	Delete what's there and paste column L from EPCMT in its place (ensure json formatting is adhered to).	BaRS	17	Daily after 4pm
29	Commit the changes and make a pull request for review.	BaRS	18	Daily after 4pm
30	Once reviewed, Merge the changes and perform a get Capability statement on one of the changed endpoints.	BaRS	19	Daily after 4pm

EPC Alignment to DoS - Weekly

Activity ID	Activity Description	Team performing	Sequence	Timeframe
32	Send a copy of EPCMT to DoS (Alex Connelly and Nichola) by 1pm on Fridays	BaRS	1	Weekly
33	Review EPCMT v DoS and DoS v EPCMT for accuracy	NECS DoS	2	Weekly
34	Alert any anomalies to BaRS team including retired, de-registrations etc.	NECS DoS	3	Weekly
35	BaRS team to make amends as required to the EPCMT and follow steps 29 - 33	BaRS	4	Weekly

Appendix 5 – Overview of ICT Service Provision

ICT service provision can be managed at different layers and either wholly in-house or by the System Supplier themselves. This is displayed in the image below; further information can be found [here](#):

Cloud computing

